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**Bridging the Gap: Utilizing Knowledge Graphs to Uncover Inconsistencies in
Social Science Theories**

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Bridging the Gap: Utilizing Knowledge Graphs to Uncover Inconsistencies in Social Science Theories

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Abstract

This paper investigates the limitations of current empirical research methods in social sciences, focusing on the conflicts and inconsistencies in social science theories. The study proposes a novel approach to address these challenges by utilizing data science methodologies, particularly the construction of knowledge graphs. By examining the relationships and patterns within existing works, the knowledge graph approach aims to identify the sources of conflicts and provide potential solutions for refining theories and improving the validity, generalizability, and reproducibility of social science research. Several case studies are used to illustrate the practical application of this approach, demonstrating its potential to shed light on inconsistencies within a specific domain, showing different types of sources of conflicts in social sciences, i.e., epistemological, ontological, methodological, and contextual conflicts. The paper concludes by discussing the implications of the findings for social science research and theory development, acknowledging the limitations of the study, and suggesting areas for future research.

Keywords: Data Science; Empirical Research Methods; Knowledge Graph; Social Science Theories.

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1 Introduction

1.1 Background and Motivation

1.1.1 Background on the Limitations of Empirical Research

One of the challenges in empirical research in empirical social sciences¹ is that it is often difficult to accurately define or separate the conditions under which social phenomena occur, making it difficult for research findings to be reproducible. In natural sciences, there exists a series of separable systems with limited and controllable variables, such as the physical systems corresponding to classical mechanics. Many linear systems exist within these, and even if a system is nonlinear, using linear models to approximate or replace nonlinear models can be successful (Chen et al., 2013). However, in social sciences, compared to natural sciences, most social phenomena we face are nonlinear and may exhibit chaotic behavior. The outcomes of such chaotic systems are sensitive to initial conditions, so if a research theory does not accurately grasp all relevant variables concerning a specific social phenomenon, differences in uncontrolled initial conditions may lead to significant variations in the corresponding empirical research results. Thus, despite our efforts to use effective experimental methods such as the control variable method, many social science experiment results still have irreproducibility (Lawless, 2017). Factors such as sample selection, experimental process, and measurement standards may all contribute to significant differences in results. Figure 1 and Table 1 illustrates the relationship between different systems.

¹ The focus in this paper is on empirical social sciences that endeavor to provide explanations. It is important to note that these disciplines necessitate the formulation of theories and generalizations to explain phenomena. This statement recognizes the existence of empirical social science studies with different objectives - such as those seeking to understand a certain aspect rather than explain it, as seen in ethnographic research. However, the discussion here does not extend to these other research orientations.

Table 1. A Comparison of Different Systems.

Sciences	Object	Properties	Approximation
Natural Sciences	Mostly linear systems	Finite and controllable variables	Usually applicable
Social Sciences	Mostly nonlinear systems	Chaotic nature	Difficult to apply

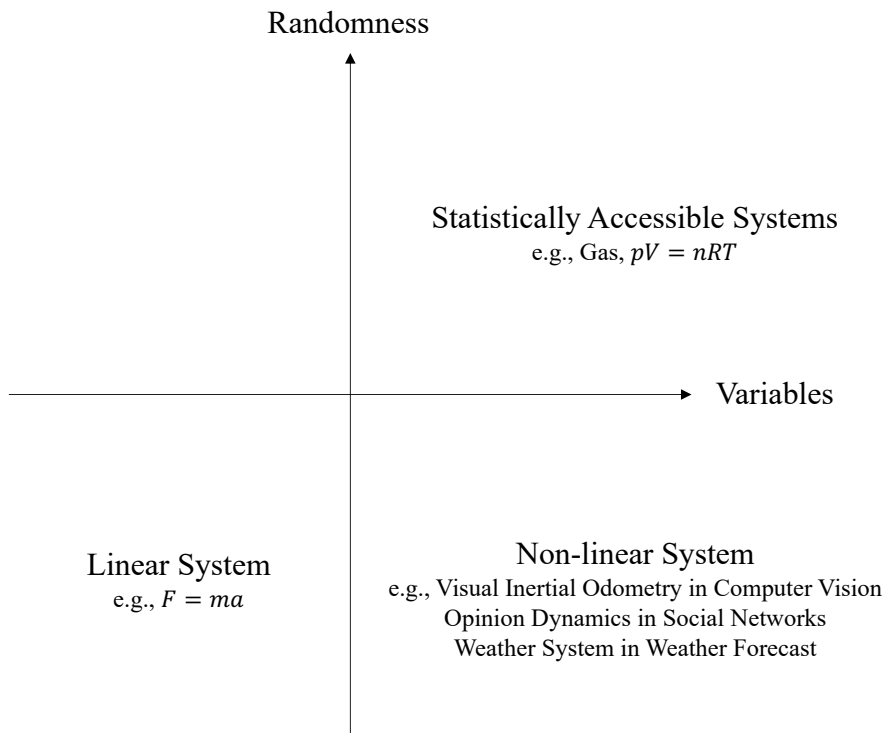


Figure 1. A Comparison of Different Systems.

This issue, to some extent, undermines the validity, universality, and repeatability of research. For example, under the theoretical background of Self-Determination Theory (SDT), Kusurkar et al. (2013) found that in the context of medical education, autonomous motivation and learning effort were not significantly positively correlated in a statistical sense, and the promotion of learning effort on GPA did not hold true for all gender groups. In contrast, other studies based on SDT theory and measurement reported opposite conclusions. For example, Boiché et al. (2008), in the context of physics education, demonstrated a positive correlation between student motivation and effort, and Schwinger, Steinmayr, & Spinath (2009) indicated

the promotion of effort on GPA in high school studies. All three studies had sample sizes exceeding 200 people and employed reasonable statistical methods. We cannot simply determine the cause of these inconsistencies logically, nor can we attribute them solely to differences in disciplinary backgrounds leading to variations in the motivation-effort-learning performance mechanism. This presents obstacles in applying relevant conclusions in real life. Table 2 and Figure 2 gives an abstraction of these papers' properties.

Table 2. An Example of Reproducibility and Consistency Problems.

Study	Context	Conclusion	Sample Size
Kusurkar et al. (2013)	Medical education	No statistically significant positive correlation between autonomous <u>motivation</u> and learning <u>effort</u> , and the promotion of learning effort on <u>GPA</u> did not hold across all gender groups.	383
Boiché et al. (2008)	Physical education	Positive correlation between student <u>motivational</u> profile and <u>effort</u> .	215
Schwinger, Steinmayr, & Spinath (2009)	High school learning	A facilitation of <u>GPA</u> by <u>effort</u> .	231

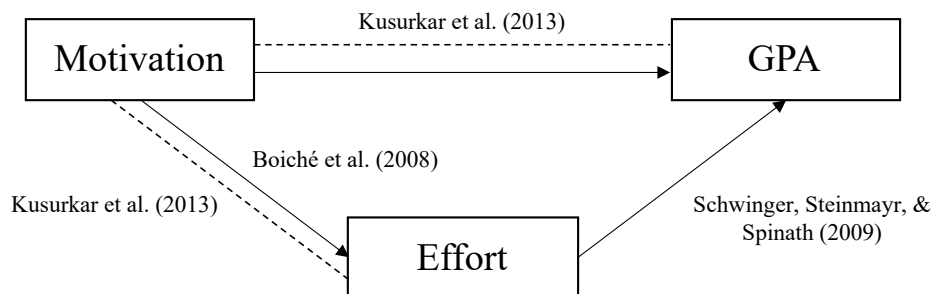


Figure 2. Different Findings on Motivation-Effort-GPA relationship.

This paper argues that this issue is not a problem with research methods in social sciences but lies in the greater difficulty of accurately defining variables, controlling environmental conditions, and measuring in social sciences compared to natural sciences. Furthermore, we lack experience in applying good mathematical and computational tools to solve corresponding problems. This phenomenon of inconsistent conclusions also provides clues that there may be issues with existing theories or that the experimental datasets may not be representative. By

researching and reconciling these contradictions, we should be able to determine under what conditions existing theories are valid and under what conditions they are invalid, thereby improving theories.

Some existing data science methods offer new tools and insights for addressing the reproducibility problem. Data science methods encompass a range of techniques that are used for handling, analyzing, and extracting meaningful information from large and complex datasets. They incorporate a fusion of disciplines, including mathematics, statistics, computer science, and domain expertise, allowing them to offer novel tools and perspectives, particularly useful in addressing issues such as reproducibility. In contrast, traditional statistical methods primarily focus on analyzing and interpreting data under a prescribed set of assumptions to draw inferences or predictions. However, they may not be well-equipped to handle the volume, velocity, and variety of data in contemporary research. Statistical methods are an integral part of data science and provide the foundation for many data analysis techniques. We will further introduce the position of traditional statistical methods as a subcategory of data science methods in the next section. This paper will first provide a comprehensive review of the current applications and limitations of data science methods in social sciences, and then, based on the discussion, further develop the discussion on how to improve the empirical research issues mentioned above using data science methods.

1.1.2 The Potential of Data Science Methodology for Addressing These Issues

Statistical Methods. The application of statistical and other data science methods in social sciences has experienced three stages, from basic statistical applications to linear causal models, and finally to modern complex modeling based on social networks and spatial data (Raftery, 2001). This classification is based on the types of data targeted by different statistical methods. As the quantity and complexity of available social data increases, new data science

methods are continually integrated into social science research, enhancing the discipline's ability to handle increasingly complex data structures. For example, modern data science applications in social sciences primarily focus on social network analysis and text sequence analysis, and some social simulation models have been proposed (Axelrod, 1997; Gandomi & Haider, 2015). These studies are now feasible mainly because the development of the internet and computer technology has made it possible to access data that could not be collected in the past, adding new variables to our research from actions and communications (González-Bailón, 2013). Regardless of the specific field, constructing and sharing reliable, large-scale raw datasets for broader analysis is a common trend across disciplines (Arbuckle et al., 2014; Davidson et al., 2019; Lazer et al., 2020). The newly data science methods are being widely applied in various fields, including social sciences, bringing new knowledge and solving problems that traditional analytical methods could not address.

Social Simulation. In terms of social simulation, some early work used agent-based models as an entry point for studying complex networks. These studies, based on social science theories, abstracted and simulated research questions, drawing preliminary conclusions and predictive hypotheses about the phenomena through computer simulation (Jackson et al., 2017). Classic examples include Pluchino, Biondo, & Rapisarda (2018), who discussed factors determining success and demonstrated the limitations of classical economics in predicting equilibrium economic systems or markets through computer simulation experiments (Epstein & Axtell, 1996). Although these studies cannot provide strict proof and are merely simulations, they can to some extent reflect the macro-level emergence of individual micro-level behaviors and enable nonlinear interactions. This type of nonlinear, network-based interaction is a characteristic of society. Traditional statistical analysis has difficulty analyzing these interactions or, in other words, integrating this type of data structure into existing analytical methods. Data science methods fill this gap.

Computational Social Science. Social network research has more application scenarios and achievements. In the hotspot issue of social contagion, interdisciplinary efforts, including social sciences, have also developed. These studies assume that the diffusion process is based on specific mechanisms and is propagated through individual interactions. Some empirical mathematical models, such as the SIR model, have been applied to the analysis of these problems, ultimately revealing relationships between contagious diseases and social factors such as family, community, misinformation, and relational networks (Zhang et al., 2020). Although these models may also have issues with overfitting, lack of representativeness, and misinterpretation, we can apply measurable tools to validate them, such as using Bayesian indirect inference to determine the best model and normalized Goodness-of-fit methods to test the predictive accuracy of the models (Troost et al., 2023). These mathematical tools can better provide evidence for the validity of empirical research.

Giddens (1986b) proposed in his structuration theory that structure has duality, and there is double involvement between individuals and institutions. We cannot study social structure or individuals in isolation; instead, we must consider both simultaneously. Current data science methods allow for the simultaneous analysis of both, which is a significant difference between computational social methods and traditional statistical applications in social science. They take into account the structure of social interactions, which is more fundamental than differences in data quantity.

Despite this, data science methods cannot replace social science research. The point of this research is to emphasize that even when using data science methods to address social science issues, the corresponding research still belongs to the realm of social sciences, rather than the realm of mathematical statistics or other natural science disciplines. The role of data science methods should be seen as a tool to support social science research and to refine and develop existing research findings. For example, from the perspective of sample size, we can

view the data collection of big data as an optimization of the difficulties and time constraints in traditional survey data collection methods. Both big data collection and survey research methods serve the same function in social science applications, which is to collect data². The functions of social network analysis and big data analysis in social science research are also similar to traditional statistical methods, providing empirical evidence and predictive hypotheses for social science research and theories. Data science methods have not become independent of traditional qualitative and quantitative social science research, forming a third category of “computer and mathematical methods.” Instead, they should be seen as a complementary tool that works alongside existing social science research methods to provide a more comprehensive understanding of social phenomena. By embracing both traditional and data science methods, social scientists can enhance their ability to study complex social interactions, networks, and systems, ultimately contributing to the advancement of the field as a whole.

1.1.3 Limitations and Boundaries of Data Science Adoption

Indeed, besides the numerous advantages discussed above, there are a series of limitations in the process of applying statistical methods to social sciences. In regard to the most common issue of hypothesis testing in statistics, Raftery (2001) proposed targeted criticism, stating that

² While both big data and traditional survey methods share the overarching objective of data collection in social sciences, there are noteworthy distinctions between the two. Surveys are specifically designed to gather targeted data, often relying on structured formats to collect individual responses. Big data, on the other hand, typically aggregates vast amounts of unstructured information from a variety of sources. Although it offers advantages in terms of volume, velocity, and variety, big data might not always yield the same precise, targeted data that surveys are designed to collect. Instead, big data often relies on proxies or indirect indicators to glean insights. Thus, while big data may optimize the time and effort required in traditional data collection, the kind of data it provides can be significantly different from that obtained through surveys. Hence, researchers should exercise discretion in selecting the appropriate data collection method, taking into account the nature of the data required and the research question at hand.

statistical laws show that p-values are more likely to reject the null hypothesis when the sample size N is quite large, even if the underlying theory is deemed meaningful. In such cases, some social science researchers might directly ignore the relevant results and insist on the correctness of the theory. Due to the complexity and difficulty of obtaining social science data, when the data is flawed or lacks representativeness, statistical results may still diverge from the theory, even if it is reasonable, negatively impacting the reproducibility of research (Sidi & Harel, 2018). Unlike natural sciences, collecting data in social sciences is more challenging. In natural sciences, we cannot ignore the differences between experimental results and theories under strictly controlled variables, while in social sciences, it is possible for data, rather than theory, to be the source of problems. However, some researchers use methods such as log-linear models to screen and validate inconsistencies between models and data (Schwarz, 1978), striving for the reliability of empirical research conclusions. Some existing work has also begun to use methods such as machine learning to validate various aspects of classical theories and research, including qualitative and quantitative conclusions (Chen et al., 2018; Grimmer, Roberts, & Stewart, 2021).

In addition, existing research in ethnography opposes "reducing ethnography to primary philosophy," which means placing the priority of theory above ethnography (Biehl, 2013). This perspective also needs to be considered when using a data science-based approach.

As Giddens points out in his work *Sociology: A Brief but Critical Introduction* (1986a), the difference between social science and natural science lies in two aspects: we cannot understand social facts in the same way as we understand objects in the natural world because social structures and social practices are interrelated, and social activities are not simply the results of cause-and-effect relationships. We might be able to obtain accurate social data, but this accuracy may not be related to reproducibility, because of a lacking simple cause-and-effect relationship in the social phenomena. This explains the first reason why he believes the

data are less reproducible than in the natural sciences, namely that structure influences practice, but practice can change structure, so the social environment does not need to be stable. This is different from the majority of systems and environments of natural sciences. The other aspect is that the meaning of social science practice and the technological applications of natural science are not entirely the same, as social activities are not mechanically determined by natural laws. Giddens' second argument lies in the question of meaning, where structure affects not only practice but also meaning, that is, the question about social science is not a mono-causal problem. Thus, it is not difficult to understand why he argues that the social sciences are less repeatable than natural sciences. Therefore, the success of statistics and approximation in social science cannot be the same as that in natural science, which is an inevitable result of differences in the objects and mechanisms studied. This difference generates the limitations of applying statistics in social sciences.

There are still many debates and controversies about what social science is and its purpose (Giddens, 1986a; Peters, 2020), which is not the issue this paper intends to discuss. Admittedly, positivism is not the only way to understand the world, and the boundaries of statistical applications may also lie here (that is, naturally, this paper emphasizes that statistics cannot solve all problems). However, this paper also needs to point out that within the scope of statistical or, more broadly, data science applications, it is crucial for empirical research in social science to pursue intersubjectivity and consistency with theory. The criticism mentioned earlier regarding the goal of control theory in social science does not deny the value of studying society as an intersubjective reality³. Whether prediction and control are the goals of social

³ In the context of this paper, the term objectivity refers to the match between the thing-in-itself, i.e., the real physical object, and the phenomenon, i.e., people's images of the object, and the term intersubjectivity means the match between phenomenon of one person and the phenomenon of another person. This paper consider the concept of society as an image in people's minds, thus we use the term intersubjective reality to emphasis that society is not a real physical object in this paper's context.

science and whether studying society objectively in some research is valuable are different questions. Computational social scientist Duncan J. Watts (2017) suggests that it is reasonable to use different theories to explain different problems, but having completely different and contradictory theories for the same problem is a key issue in modern social science research. This widespread phenomenon makes it difficult to apply theories to solving practical problems. His criticism believes that the current social science trend of encouraging the development of new theories rather than verifying or refuting previously published theories is the cause of the inconsistency issue and has not been resolved with the advancement of computational social science. This is an urgent problem that social science research needs to address. From the perspective of scientific development, it is meaningful for theories to be generalizable and practical.

1.2 Research Purpose

As a result, this paper focuses on the reproducibility issue in social science empirical research and the related practicality problems. Some existing work in computational social science has shown that data science methods, represented by big data analytics, enable classical social science theories to be validated and practically applied in various scenarios (Watts, 2017; Lazer et al., 2009; Lazer & Radford, 2017; Rahwan et al., 2022). This paper will first conduct an in-depth investigation of the causes of the reproducibility issue in social science through a literature review. Secondly, the paper will use case analysis to examine the patterns of applying data science methods to social science empirical research. Finally, the paper will summarize these patterns and provide general recommendations on how to use data science methods to improve the performance of social science empirical research in terms of reproducibility and consistency. In this sense, the following research questions are posed in this paper:

RQ1. What are the practical drawbacks of current quantitative empirical research methods in social sciences?

RQ2. Can data science methods improve the above problems?

1.3 Structure of the Dissertation

This paper is organized as follows: Section 2 provides an overview of the phenomena of theoretical conflict and inconsistency in social science and introduces the use of knowledge graphs in social science. Section 3 describes the methods, including data collection and processing, and the construction of knowledge graphs. Section 4 presents the results in the form of case studies. Section 5 discusses the results and explores methodological issues. Finally, section 6 concludes the paper and presents limitations and directions for further research.

2 Literature Review

2.1 Overview of Conflicting Theories

2.1.1 Reproducibility Issues in All Sciences

Before discussing the issues of conflicts and inconsistencies in theories, we must first clarify why consistency and reproducibility are important. The significance of reproducibility is emphasized in both quantitative and qualitative research (Clonts, 1992; Sui & Kedron, 2021). The importance of theoretical coherence and reproducibility cannot be overstated in the pursuit of a deeper understanding of social phenomena and in ensuring the reliability of scientific outcomes. Theoretical coherence pertains to the logical integration and consistency of theoretical components, claims, and assumptions within individual studies and across the field. This consistency enables researchers to expand upon previous work, fostering the development of a unified body of knowledge (Resnik & Shamoo, 2017). Some specific examples raise the tension between the coherence of social science theories and pragmatism (Antona & Babin, 2001). Conversely, reproducibility concerns the capacity of independent researchers to either duplicate a study's findings using identical methods and data or achieve analogous results with new data under similar conditions. Reproducibility bolsters the trustworthiness and applicability of research outcomes, as it reduces the impact of chance errors, biases, or other confounding variables that may be present in a single study. Additionally, reproducibility promotes openness and cooperation within the scientific community by encouraging the sharing of methodologies, data, and results to enable replication efforts. In essence, theoretical coherence and reproducibility provide a solid and rigorous basis for social science research, ultimately supporting the creation of effective solutions, policies, and practices to tackle complex social challenges. Thus, Popper (2005) believes that reproducibility should be one of the hallmarks of good science as a standard. And this paper also argues that reproducibility is an important factor for a science to be scientific.

Although the experimental conditions of empirical research in social sciences are limited, reproducibility issues are not unique to social sciences. A series of existing critiques point out that reproducibility problems exist in both natural and social sciences, as well as in qualitative and quantitative research (Howick et al., 2022; Ioannidis, 2012; Ioannidis & Doucouliagos, 2013; Smaldino & McElreath, 2016). The lack of reproducibility is a common issue in science, and the difference between social and natural sciences does not lie in reproducibility.

Based on these facts, this paper argues that we can use more data science tools to support reproducible empirical research and measure reproducibility.

2.1.2 Comparison between Social Sciences and Natural Sciences

Quantitative Research. A significant difference between social sciences and natural sciences in empirical research lies in the difficulty of experimentation (Knorr-Cetina, 1981). As mentioned earlier, it is often challenging to accurately define the conditions and boundaries of social phenomena. The emergence mechanisms of society make it impossible to represent some given social mechanisms with simplistic mechanical descriptions of individuals and their interactions (Sawyer, 2004). Therefore, the success of reductionism in natural sciences cannot be fully applied to social sciences (Bickle, 2007; Louth, 2011; Verschuren, 2001), and the scope of mechanical explanations may be limited (Sawyer, 2004). These characteristics of social science research subjects objectively pose challenges for empirical research in social sciences.

Theories serve as intermediaries for variables (Baranowski et al., 1997). A theory indicates all relevant variables involved and variables assumed to be unrelated to the study and thus excluded. In natural sciences, these variables are usually more clearly defined, measurable, and have definite boundaries, making it easier to identify and control them. In contrast, accidental factors in environmental conditions are generally less common in natural science

research compared to social science research. Therefore, the main differences between the two sciences in experimentation lie in the difficulty of identifying variables, the ability to control environmental conditions, and the validity of measuring variables. Of course, natural sciences are not perfect in controlling variables either. Both natural and social sciences have difficulties in determining all variables and controlling them (Müller & Roeder, 1997; Sturman, Sturman, & Sturman, 2022), so the difficulty of controlling variables is not an essential difference between the two disciplines. Nevertheless, empirical research in social sciences remains more challenging compared to natural sciences.

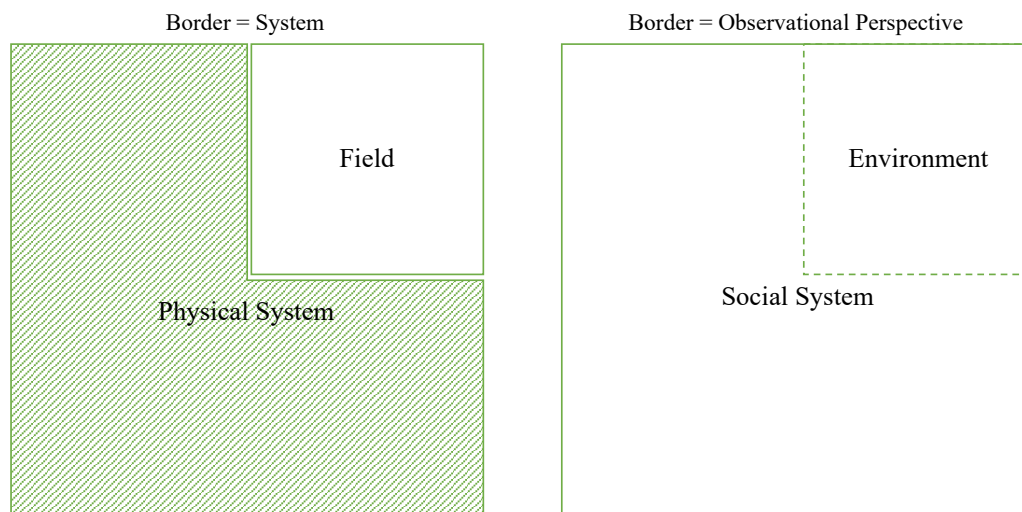


Figure 3. Why Reductionism Does Not Apply to All Problems.

On the other hand, the difficulty of controlling environmental conditions in social sciences makes it easier to find flaws in theories. As mentioned earlier, we can easily find contradictory conclusions in some psychological studies, revealing the limitations of theories under specific conditions. This is harder to detect in natural sciences because they can effectively set excluded variables as constants, which may lead to limited or erroneous laws by excluding factors that actually influence the research problem.

Apart from experimental issues, as Giddens (1986a) points out, social sciences do not exclusively discuss objects of the natural world. For the non-objective world, social sciences require different structures to address subjective issues, that is, qualitative research. Qualitative

research is about the study of meaning (Silverman, 1998). The fundamental difference between social sciences and natural sciences, as well as between qualitative and quantitative research, lies in the nature of their research subjects. In natural sciences and quantitative research, the issue of meaning is not discussed. Conversely, qualitative research attempts to identify the mental images of objects (Kant's things-in-themselves) and ideas (phenomena) (Schulze, 1792), giving meaning to objects and ideas without the need to accurately describe them. In social sciences, the same concept may have different meanings, while the laws sought by quantitative research are usually unique under certain conditions. Therefore, qualitative and quantitative research methods, frameworks, and evaluation criteria are different. We need to accurately distinguish between them in discussions.

Qualitative Research. The purpose of qualitative research is to understand and interpret individual experiences (Pope & Mays, 1995), which is a non-experimental observation. We cannot evaluate qualitative research with the criteria used for measuring quantitative research, as qualitative observations are clues and possibilities, but they need not and cannot be entirely objective. It can be argued that qualitative and quantitative research in social sciences each fulfills the function of proposing and testing theories, and they are independent and irreplaceable. Based on the foundation of social science theories, data science methods can support both the proposition and testing of theories, but they cannot be used without the context of theoretical backgrounds. From the perspective of control theory, opposing the substitution of social science with control theory methods means that we should not only focus on the relationships between statistical data but also interpret them using qualitative methods to avoid fallacies. Qualitative methods make data results interpretable (Mills, 2018; Mills, 2019).

The advantage of qualitative research in social sciences when proposing theories lies in the fact that, compared to natural sciences, the connections between factors in social sciences are looser, and the structure of social science subjects is more relaxed. Using qualitative

research to interpret phenomena and propose research hypotheses is of great significance. Taking a study that will be discussed in detail later as an example, Acemoglu, Johnson, & Robinson (2001) suggested that colonial mortality rates could serve as a good empirical tool for measuring institutional effects. Although we have a wealth of data on the colonial period, it is difficult to accurately define concepts and variables from the massive amount of data and apply them to empirical research. On the other hand, quantitative research methods like data science lack interpretive capabilities, such as the shortcomings of machine learning in interpretability and transparency (Du, Liu, & Hu, 2019; Malik, 2020). The results they propose may be mathematically correct, but for practical applications, we still need meaningful theories to explain them. Thus, qualitative research plays an essential functional role in proposing theories and interpretations, because quantitative research methods, including data science cannot give the data meaning.

2.1.3 The Nature of Conflicts

Building upon the aforementioned discussion, we examine the nature of theoretical conflicts and the limitations of existing empirical research methods from both quantitative and qualitative perspectives. First, we discuss the issues in quantitative research in social and natural sciences. As previously mentioned, the current problems in social science research arise not only from ethical constraints and insufficient application of mathematical tools, but also from the fact that social systems are nonlinear and complex. In complex systems, the behavior of entities cannot be simply summarized using statistical methods (without exhibiting a unified behavioral pattern), and it is difficult to analyze the interaction patterns among all entities using analytical methods due to the sheer number of elements and relationships involved (Weinberg, 1975). Social systems exemplify such complexity, and some individual behaviors and differences lose sociological significance when averaged, or lose their individual nature,

rendering direct application of statistical methods unsuitable for analysis. The numerous factors influencing society make it difficult to obtain patterns and data for all interactions, thus impeding our understanding of society through analysis of all relationships. The inherent properties of social systems make it difficult to use traditional statistics to understand causal relationships in most nonlinear social phenomena (Ladyman, Lambert, & Wiesner, 2013). In addition to complexity, social systems also exhibit chaotic behavior. In some nonlinear systems that produce chaotic phenomena, small perturbations can cause significant differences in outcomes, and these perturbations may be beyond the scope of experimental observation or considered irrelevant by research assumptions (Lorenz & Haman, 1996). Meteorological systems are a typical example of chaotic systems, posing challenges for explanation and prediction. This issue is not only difficult within the scope of social sciences but also presents a significant challenge in natural sciences.

To address these challenges, which traditional statistical methods struggle to effectively resolve, Watts (2017) points out that increasing the amount of empirical research data can help establish broader and more effective theories and test existing theories. However, he also notes that social sciences still have a gap in forming intrinsically consistent models. Macro-social experiments struggle to establish appropriate causal relationships for social phenomena. We need to introduce new methods to refine empirical research patterns in social sciences, attempting to find models and theories with finer granularity, greater interpretability, and stronger predictive power.

To tackle these issues, some literature review methods based on statistics have been employed. For example, a meta-analysis spanning 40 years was introduced to summarize whether external incentives would affect intrinsic motivation (Cerasoli, Nicklin, & Ford, 2014), and it was found that the two were not necessarily opposed. However, the paper also incisively points out that meta-analysis cannot guarantee the accuracy of conclusions, and subjectivity

still exists in the coding process. The paper also suggests that meta-analysis is ineffective for nonlinear relationships, and we cannot assume that the relationships between the elements under study are linear. Thus, although this research effectively resolves some contradictions in traditional research conclusions and provides clues for further research into intricate causal relationships, it still cannot accurately elucidate the coexistence mechanism between incentives and intrinsic motivation.

In contrast to the above techniques based on simple statistics, biological researchers attempt to understand individual behavior principles from the perspective of neural mechanisms. Murayama et al. (2010) used functional magnetic resonance imaging to analyze the neural correlates of incentive effects on intrinsic motivation, confirming the existence of “undermining effect” and proposing the neural basis for it (the integration of external incentive value and internal task value). This conclusion offers a more detailed explanation of the mechanism and improves reproducibility.

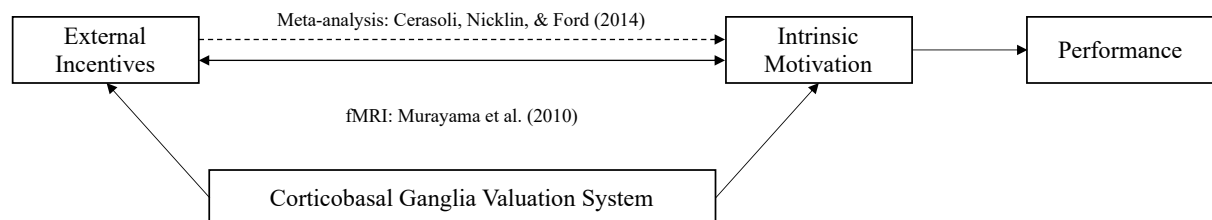


Figure 4. Research Diagram of Undermining Effect.

As some criticisms of the limitations of falsificationism point out, some theories may still have value in terms of thought reference and explaining a portion of phenomena even if they could be falsified at the beginning of their development (Lakatos, 1976). Social science efforts in macro-social experiments do indeed provide insightful explanations for some social phenomena, and corresponding theories possess predictive capabilities for some linear system issues, which is indisputable. However, the development of theories is always accompanied by a process of falsification. With advances in computational power and progress in addressing nonlinear problems, we should be able to identify methodological issues in existing research

and improve or even resolve them by integrating a broader range of research methods based on their characteristics.

Secondly, in terms of qualitative research in social sciences, it is true that different perspectives will lead to varying interpretations of the same subject, which is acceptable. However, not all such disagreements are beneficial to social and social science development. Clunas (1999), in discussing cultural backgrounds, presents different reasons for the divergence of qualitative arguments among theories from other perspectives. He believes that disagreements are beneficial, but there should be a relatively consistent interpretation for these disagreements, which is related to the various perspectives mentioned earlier. He points out that research is limited by region and era, and although similar issues may be discussed, different conclusions will be reached based on different standpoints, assumptions, and backgrounds. For example, Clunas's example illustrates that interpreting Chinese art from a Western perspective is inherently difficult, and vice versa. To make disagreements more consistent, a common foundation and set of questions are needed for research. This interdisciplinary research approach can provide insights for solving problems.

In addition to these general discussions on the limitations of existing empirical research methods, more in-depth case analyses need to be conducted to accurately understand the problem structure of different empirical research methods and seek potential data science improvement solutions based on these analyses.

2.1.4 Examples of Conflicting Theories

Various domains of social science research exhibit several examples of conflicting theories. Some notable instances include:

In economics, the Keynesian and classical schools of thought present conflicting perspectives on the role of government intervention in the economy. Keynesian economists

argue that government intervention is necessary to stabilize economic fluctuations and promote growth (Keynes, 1937), whereas classical economists believe that free markets are self-regulating, and government intervention can be counterproductive (Smith, 1776).

In psychology, the nature versus nurture debate epitomizes a long-standing conflict between theories that emphasize the role of genetic factors in shaping human behavior (Plomin, 2018) and those that concentrate on environmental influences (Bronfenbrenner, 1979). This conflict has implications for numerous psychological phenomena, including intelligence, personality, and mental health.

In sociology, structural-functionalism and conflict theories propose competing explanations for the organization and functioning of societies. Structural-functionalism posits that societies are structured to maintain stability and cohesion, with each component serving a specific function (Parsons, 1952). Conversely, conflict theory contends that societies are characterized by power struggles and competition between different social groups, leading to inequality and social change (Marx & Engels, 1848).

In political science, realist and liberal theories of international relations offer distinct perspectives on the nature of state behavior and the prospects for global cooperation. Realist theories emphasize the role of power and self-interest in shaping state actions (Morgenthau, 1985), while liberal theories focus on the potential for cooperation and the importance of international institutions (Keohane & Nye, 1977).

2.1.5 Key Factors Affecting the Reliability of Scientific Findings

The reliability of scientific findings is a crucial aspect of research, ensuring that the outcomes and conclusions drawn from studies are trustworthy and provide a solid foundation for further exploration and application. Several key factors affect the reliability of scientific

findings, including reproducibility, sample size, research context, and variables (Ioannidis, 2005).

Reproducibility. Reproducibility is paramount, as it demonstrates the consistency of results when the same study is conducted under similar conditions (Baker, 2016). A lack of reproducibility can lead to false conclusions and hinder scientific progress. As what we have reviewed, reproducibility is more challenging in social sciences compared to natural sciences. In natural sciences, experimental conditions can often be precisely controlled, and variables can be isolated, making it easier to replicate studies and achieve consistent results. In contrast, social sciences deal with complex and dynamic phenomena that are influenced by numerous factors, many of which cannot be controlled or isolated. This makes it difficult to replicate studies and achieve consistent findings, leading to a higher degree of uncertainty and variability in social science research.

Sample Size. Sample size is another crucial factor, as it directly impacts the statistical power and generalizability of findings (Button et al., 2013). Inadequate sample sizes can result in misleading conclusions, whereas larger samples are more likely to produce accurate, representative outcomes. In natural sciences, researchers often have access to large datasets or can conduct experiments with a large number of subjects or iterations, which increases the statistical power and reduces the likelihood of spurious findings. In social sciences, however, obtaining large sample sizes can be more challenging due to practical constraints, such as limited access to populations, time-consuming data collection processes, and ethical considerations. Smaller sample sizes in social science research can increase the likelihood of sampling bias and reduce the generalizability of findings, contributing to the inconsistencies and conflicts observed in the literature as we shown above.

Research Context and Variables. Research context refers to the specific conditions under which a study is conducted, such as the study's population, setting, and timing. The

context can significantly influence the reliability of scientific findings, as it determines the extent to which the results can be generalized to other populations or settings (Merton, 1968). The variables involved in a study, both independent and dependent, also impact the reliability of scientific findings. The careful selection, measurement, and control of variables are essential to ensure that the observed relationships accurately represent the phenomena being studied (Bollen, 1989). In natural sciences, research contexts are often well-defined and relatively consistent across studies, allowing for more straightforward comparisons and generalizations. For example, the laws of physics are consistent across time and space, and biological processes generally follow the same principles regardless of the specific organism or environment. In contrast, social sciences deal with human behavior and social phenomena that are highly context-dependent, influenced by factors such as culture, history, and individual differences. This makes it challenging to establish universal principles or laws that can be applied across diverse settings. As a result, findings from social science research are often more sensitive to the specific context in which the research was conducted, making it difficult to compare and generalize results across studies. And thus, some arguments argue that predictions in social science are generally unreliable (Little, 1993). This context-dependence contributes to the inconsistencies and conflicts observed in social science theories and highlights the need for innovative approaches, such as the knowledge graph methodology, to better understand and reconcile these differences.

To sum up, the reliability of scientific findings is affected by a complex interplay of factors, including reproducibility, sample size, research context, and variables. By addressing these factors and ensuring rigorous methodology, researchers can enhance the reliability and trustworthiness of their findings, ultimately contributing to more robust and impactful scientific progress. These factors will be studied in further case studies.

2.2 Reconciling Inconsistencies in Social Science Research

Reconciling inconsistencies in social science research is crucial for establishing a coherent understanding of social phenomena and promoting the cumulative progress of knowledge in the field. Various approaches have been developed to address conflicting theories and empirical findings, which can be broadly categorized into traditional approaches and data science approaches.

2.2.1 Traditional Approaches

Traditional approaches to reconciling inconsistencies typically involve re-examining the underlying assumptions and research methods, as well as considering alternative explanations and theoretical frameworks. These approaches may include the use of meta-analysis, critical reviews of the literature, and the application of robustness checks to test the sensitivity of research findings to changes in methodology and data (Stanley & Doucouliagos, 2012).

Meta-analysis is a quantitative method that combines results from multiple studies to derive a more reliable and generalizable estimate of effect sizes, while accounting for potential biases and heterogeneity across studies (Borenstein et al., 2021). By synthesizing evidence from a large number of studies, meta-analysis can help to reconcile conflicting findings and identify factors that contribute to discrepancies, such as differences in sample characteristics, study designs, and measurement methods. However, as we stated before, meta-analysis cannot guarantee the accuracy of conclusions and is ineffective for nonlinear relationships. It is very hard for meta-analysis to provide us new perspectives out of the methodologies from the original papers used in meta-analysis.

Critical reviews of the literature involve a systematic evaluation of the quality and consistency of evidence on a given topic, taking into account the methodological rigor, theoretical coherence, and empirical support for competing explanations. This process can help

to identify gaps in knowledge, clarify the conditions under which different theories apply, and highlight areas where further research is needed (Petticrew & Roberts, 2008).

Robustness checks, which involve testing the sensitivity of research findings to changes in methodology and data, can help to assess the reliability and generalizability of results, as well as to identify potential sources of bias and confounding (Leamer, 1998). By demonstrating the stability of research findings across different specifications and sub-samples, robustness checks can contribute to the resolution of inconsistencies and increase confidence in the validity of theoretical claims.

2.2.2 Data Science Approaches

Data science approaches to reconciling inconsistencies in social science research involve the application of advanced statistical methods, computational techniques, and large-scale data sets to test and refine theories, as well as to discover new patterns and relationships in the data. These approaches may include machine learning, network analysis, and natural experiments, among others (Salganik, 2019).

Machine learning techniques can be used to model complex, non-linear relationships between variables, allowing researchers to identify previously unrecognized patterns and interactions that may help to reconcile conflicting theories (Bishop & Nasrabadi, 2006). By exploring a wide range of possible models and selecting the best-fitting ones based on predictive accuracy and model complexity, machine learning can contribute to the development of more nuanced and robust theoretical explanations.

Network analysis provides a set of tools for studying the structure and dynamics of social systems, allowing researchers to identify patterns of relationships and dependencies that may underlie observed inconsistencies (Wasserman & Faust, 1994). By examining the position and

role of different actors and variables within a network, researchers can gain insights into the mechanisms that drive social phenomena and test competing explanations for observed patterns.

Natural experiments, which involve the identification of exogenous shocks or interventions that affect social systems in ways that are unrelated to the outcome of interest, can help to establish causal relationships and reconcile conflicting theories by providing quasi-experimental evidence (Dunning, 2012). By exploiting variation in the treatment and control groups that arises from naturally occurring events, natural experiments can help to address issues of endogeneity and omitted variable bias that may contribute to inconsistencies in the literature.

These data science approaches, combined with traditional methods, can provide a comprehensive toolkit for addressing inconsistencies in social science research. By integrating diverse sources of evidence and leveraging the power of advanced computational techniques, researchers can work towards reconciling conflicting theories and enhancing the robustness and credibility of social science knowledge. Ultimately, this can lead to a more accurate and nuanced understanding of complex social phenomena, informing the development of effective policies and interventions to address pressing societal challenges. In this paper, we aim to use knowledge graphs to address inconsistency problems.

2.3 Introduction to Knowledge Graphs and Their Applications

One promising data science approach to address inconsistency problems in social science research is the use of knowledge graphs. Knowledge graphs are a type of structured data representation that can model complex relationships between entities, attributes, and events in a way that is both human-readable and machine-processable (Ehrlinger & Wöß, 2016). By extracting and organizing information from diverse sources, such as research articles, databases, and expert knowledge, knowledge graphs can provide a comprehensive and dynamic picture

of the research landscape, facilitating the identification and resolution of inconsistencies between different theories.

2.3.1 Definition and Components of Knowledge Graphs

A knowledge graph is a structured, semantic representation of knowledge that models entities, their attributes, and the relationships between them (Ehrlinger & Wöß, 2016). Knowledge graphs provide a flexible and expressive way to represent complex information and enable machines to understand, reason, and infer new knowledge from the graph.

The primary components of knowledge graphs are:

Entities. These are the main objects or concepts represented in the graph. Entities can be people, organizations, events, or any other real-world or abstract concept relevant to the domain of interest.

Attributes. These describe the properties or characteristics of entities. Attributes can be quantitative (e.g., age, height) or qualitative (e.g., nationality).

Relationships. These represent the connections between entities, defining how they are related to one another. Relationships can be directed or undirected, and they can have different types or strengths depending on the nature of the connection.

Ontology. This defines the structure and semantics of the knowledge graph, specifying the types of entities, attributes, and relationships that can be represented, as well as the rules and constraints governing their use. An ontology provides a shared vocabulary and conceptual framework for organizing and integrating information from diverse sources.

In addition to these core components, knowledge graphs can also incorporate other types of information, such as provenance metadata (e.g., source, author, timestamp) and confidence scores or weights for different pieces of information, reflecting the degree of certainty or consensus associated with the data (Paulheim, 2017).

2.3.2 How Knowledge Graphs Identify and Handle Inconsistencies Between Theories

The construction of knowledge graphs involves several steps, including information extraction, entity disambiguation, and relationship identification (Paulheim, 2017). Natural language processing (NLP) techniques can be employed to extract relevant concepts and relationships from research articles, while machine learning algorithms can be used to disambiguate and classify entities based on their attributes and contexts. Once the knowledge graph is constructed, researchers can apply graph analysis techniques to identify key patterns, clusters, and discrepancies within the data, shedding light on areas of consensus and disagreement between different theories.

Several recent studies have demonstrated the potential of knowledge graphs for addressing inconsistency problems in social science research. Knowledge graphs enable the representation of complex relationships between various entities, facilitating the integration of heterogeneous data sources and the discovery of hidden patterns and insights (Paulheim, 2017). For instance, some current works demonstrated the application of knowledge graphs in predicting poverty levels across different countries by integrating diverse data sources, such as GDP, demographic indicators, and geographic factors (Song et al., 2023; Stamou, 2022; Yun et al., 2019). Similarly, Dou et al. (2018) leveraged knowledge graphs to model and analyze the relationships between cultural heritage objects, enriching cultural heritage research by unearthing new connections and insights.

In the domain of political science, Xia and Gu (2019) employed a knowledge graph-based approach to analyze global terrorism events, revealing underlying patterns and trends that could aid in understanding the complex nature of terrorism. Furthermore, Chen et al. (2018) proposed the use of knowledge graphs for linking open data in education, facilitating a comprehensive understanding of the factors that influence educational outcomes.

These examples underscore the value of knowledge graphs in integrating diverse data sources and addressing inconsistency problems in social science research, as they offer a unified framework for representing and analyzing complex relationships between various factors and theories. By leveraging knowledge graphs, researchers can uncover novel insights and reconcile inconsistencies, contributing to the advancement of social science research.

In general, knowledge graphs can be particularly helpful in addressing inconsistencies in social science research by three aspects.

Enhancing Transparency. By providing a clear, structured representation of the relationships between different theories, knowledge graphs can help researchers and policymakers better understand the assumptions, evidence, and logical connections underlying competing explanations for social phenomena.

Facilitating Hypothesis Testing. Knowledge graphs can be used to generate specific, testable hypotheses about the conditions under which different theories are more or less applicable, guiding the design of targeted empirical studies and natural experiments to evaluate these predictions.

Promoting Interdisciplinary Collaboration. By integrating insights from multiple disciplines and research traditions, knowledge graphs can serve as a common language and framework for interdisciplinary collaboration, fostering the exchange of ideas and the development of novel, integrative theories that can overcome the limitations of single-discipline approaches.

Overall, these existing preliminary works positively show the potential of the knowledge graph approach in attempting to deal with inconsistencies in social science research and are the reason for the selection of the approach in this paper.

3 Methodology

3.1 Architecture

This study uses a case study approach to find several different problem structures in current social science research, represent their characteristics, and explore the causes of these problems in the context of social science theory and data science explanations. This paper proposes a knowledge graph-based approach to discover, visualize, and summarize these problem structures. Figure 5 presents the framework of this knowledge graph approach. There is some existing work that also uses the knowledge mapping approach to study social science hypotheses or specific social science problems (de Haan, Tidli, & Beek, 2021).

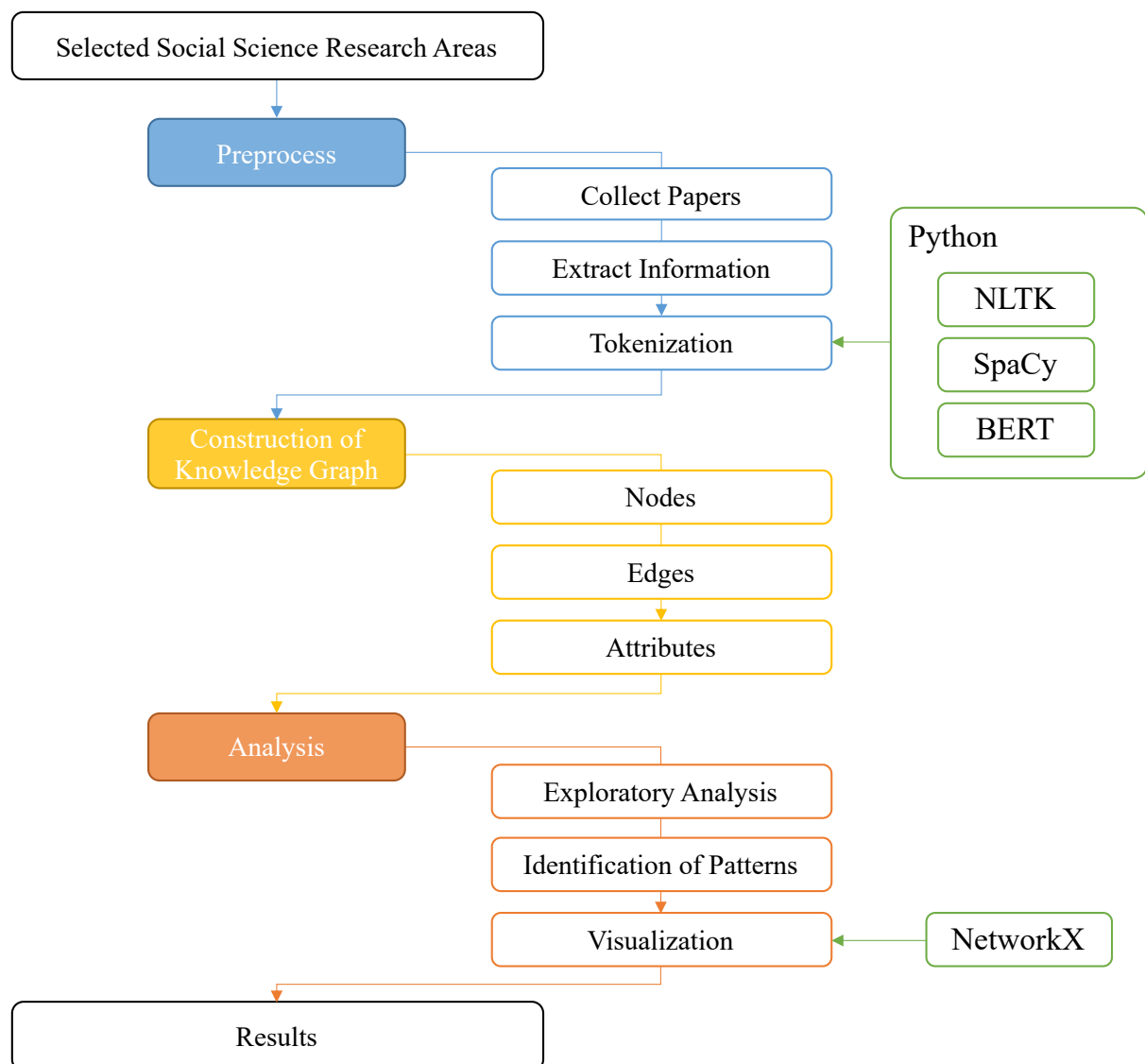


Figure 5. Knowledge Graph Framework.

The main part of program code used in this paper is shown in Appendix. We use *Python* to realize the program.

3.2 Data Collection and Preprocessing

This section describes the process of data collection and preprocessing, which involves identifying relevant social science research areas with conflicting theories, extracting pertinent information from selected studies, and preprocessing the data for constructing a knowledge graph.

3.2.1 Identification of Relevant Social Science Research Areas with Conflicting Theories

The first step in our data collection process involves deciding on specific social science fields that exhibit conflicting theories. To identify suitable research areas, we focus on theories with substantial empirical evidence supporting their validity, yet demonstrating inconsistencies across studies. By selecting well-established, yet conflicting theories, we aim to investigate the potential of knowledge graphs in reconciling these inconsistencies and advancing our understanding of these research areas.

This study utilized the Python program to search and select research papers, with articles sourced from *Web of Science* and *Scopus*, two commonly used academic databases. The keywords used to search for these academic papers included social science, natural science, quantitative, qualitative, control variables, reductionism, holism, data, big data, empirical research, reproducibility, replicability, and/or inconsistency. The exclusion criteria for the program were as follows: (1) empirical studies using closed-ended survey questions as data collection methods, as question design may be influenced by potential designer biases; (2) research in subfields where reproducibility is not an issue, or where generalizability is not required, such as studies in ideological research (Larsson, 2009).

3.2.2 Extraction of Relevant Information

Once the research areas have been identified, the next step is to extract relevant information from the selected studies. This includes information on study design, sample characteristics, measurements, and findings, as well as any other contextual factors that may contribute to the observed inconsistencies. A careful and systematic extraction of this information is crucial for building a comprehensive knowledge graph that can accurately represent the relationships and patterns across the selected studies. As stated before, sample size, research context, and variables are considered in our research.

3.2.3 Data Preprocessing

The final step in the data collection process is preprocessing the extracted information to convert it into a structured format suitable for constructing the knowledge graph. In this paper, we employ *Python* libraries, such as *NLTK* (Natural Language Toolkit) and *spaCy*, to preprocess the data into tokens. This involves cleaning the text data, removing any irrelevant information, and tokenizing the text into words or phrases that can be used as building blocks for the knowledge graph. This structured and preprocessed data serves as the foundation for constructing a knowledge graph that effectively represents the relationships between various entities and factors in the selected research areas.

3.3 Knowledge Graphs Approach

This section discusses the process of constructing a knowledge graph to represent conflicting theories in social science research. Knowledge graphs are composed of nodes, edges, and attributes, which together capture the entities, relationships, and properties within a particular domain.

3.3.1 Components of Knowledge Graph

Nodes. Nodes represent entities such as theories, studies, variables, and sample characteristics in the context of social science research (Paulheim, 2017). These nodes serve as the building blocks for representing the complex relationships between different aspects of the selected research areas.

Edges. Edges represent relationships among the entities, such as 'supports', 'contradicts', 'measures', 'examines', 'is_part_of', etc. (Croitoru et al., 2018). These relationships enable the knowledge graph to capture the interactions and dependencies between various entities in the research domain.

Attributes. Attributes describe properties of the entities, such as study design, sample size, measurement instruments, effect sizes, etc. (Hogan et al., 2021). These attributes provide additional context and details about the entities, which can be useful for understanding the factors contributing to the observed inconsistencies in the research findings.

3.3.2 Construction of the Knowledge Graph

Node Creation. The first step in constructing the knowledge graph is to create nodes for each identified entity, ensuring that each node has a unique identifier (Paulheim, 2017). This process involves systematically converting the extracted information from the studies into nodes representing the various entities in the research domain.

Edge Creation. The next step is to establish edges between related nodes based on the relationships identified during data extraction and preprocessing (Sheth et al., 2018). This process involves connecting the nodes in the knowledge graph to represent the complex web of relationships between different entities in the research areas.

Attribute Assignment. Finally, attributes are assigned to the nodes and edges based on the properties extracted from the studies (Hogan et al., 2020). This step involves enriching the

knowledge graph with additional information about the entities and relationships, providing a comprehensive representation of the research domain.

3.4 Analysis of the Knowledge Graph

This section discusses the analysis of the constructed knowledge graph to better understand the inconsistencies in social science theories. The analysis process includes exploratory analysis, identification of patterns, hypothesis generation, hypothesis testing, and validation and robustness checks.

3.4.1 Exploratory Analysis

The exploratory analysis involves generating descriptive statistics and visualizations of the knowledge graph to understand its overall structure and properties. This step enables researchers to examine the distribution of nodes, edges, and attributes, as well as the connectedness and centrality of the graph components.

3.4.2 Identification of Patterns

To identify patterns in the conflicting theories, graph analytics techniques, such as centrality analysis and clustering, are employed. Centrality analysis reveals the most influential nodes in the graph, which can represent key theories, researchers, or variables. Clustering algorithms, such as Girvan-Newman algorithm (Levens, 2021), help to identify communities within the graph, potentially uncovering relationships between conflicting theories and their underlying assumptions or methodologies.

3.4.3 Hypothesis Generation and Testing

Based on the identified patterns, researchers can develop hypotheses regarding the factors contributing to the observed inconsistencies. These hypotheses may involve the role of specific variables, sample characteristics, or methodological choices in driving the differences between conflicting theories. Formulating these hypotheses is crucial for guiding the subsequent testing and validation processes.

3.4.4 Further Functions to be Implemented

As further work, hypothesis testing can be conducted using statistical techniques, such as regression analysis or network simulations. These methods allow researchers to assess the validity of the proposed hypotheses by examining the relationships between the identified factors and the inconsistencies in the conflicting theories. The results of the hypothesis testing can provide valuable insights into the underlying causes of the observed inconsistencies and inform future research directions.

Also, to ensure the reliability of the analysis results, cross-validation and sensitivity analysis can be performed as further work. Cross-validation involves splitting the dataset into multiple subsets and testing the hypotheses on each subset independently, while sensitivity analysis assesses the robustness of the results by varying key parameters and assumptions. These validation techniques can help to confirm the generalizability and stability of the findings, contributing to a more comprehensive understanding of the inconsistencies in social science theories.

4 Results

4.1 Case Study 1: Contextual Conflicts and Research Context Factor: The Relationship between Income Inequality and Social Trust in Politics and Economics

One intriguing example of contextual conflicts in social sciences is the relationship between income inequality and social trust. The literature on this topic has produced mixed findings, with some studies indicating a negative relationship between income inequality and social trust, while others argue that the relationship is contingent on factors such as a country's level of economic development and the presence of robust social welfare policies. Figure 6 shows the findings in different literatures. Considering the limited scope of knowledge mapping, we select the most representative conflicts to present, and the same applies to the case studies afterwards. 30 to 60 relevant studies are selected to be analyzed for each case study.

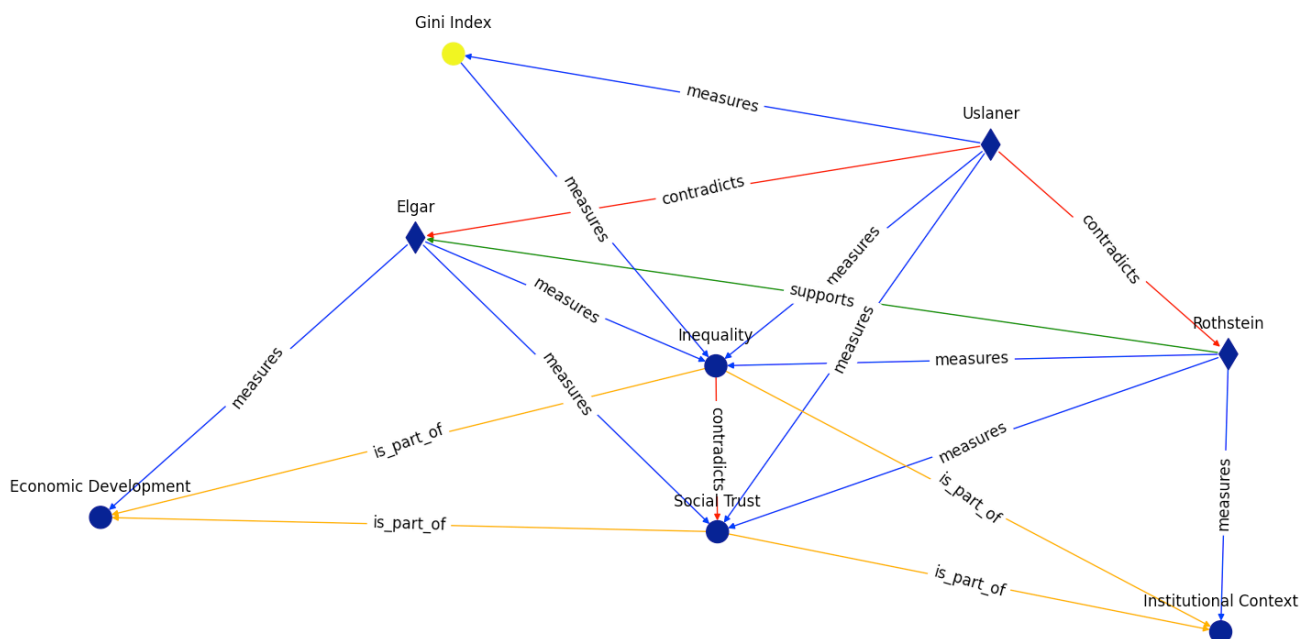


Figure 6. Knowledge Graph: The Relationship between Income Inequality and Social Trust.

Circle shape represents variables or theories, and diamond shape represents literature. The same color of the nodes means the nodes are in the same cluster identified by the program.

The same applies to the case studies afterwards.

Uslaner (2002) presents evidence of a negative relationship between income inequality and social trust, arguing that higher levels of inequality erode trust in societies. In this view, unequal societies create a sense of unfairness and insecurity, which undermines the foundation of trust among individuals. The study uses cross-national data from the World Values Survey to demonstrate a negative correlation between income inequality (measured using the Gini coefficient) and social trust.

On the other hand, Rothstein and Uslaner (2005) argue that the relationship between income inequality and social trust is not straightforward. They posit that the relationship between these two variables is mediated by the quality of a country's institutions, particularly its social welfare policies. According to Rothstein and Uslaner, countries with strong welfare states and low corruption levels exhibit higher levels of social trust, even in the presence of income inequality. This suggests that the negative relationship between income inequality and social trust may be contingent on the institutional context.

A study by Elgar and Aitken (2011) provides evidence that the relationship between income inequality and social trust may be influenced by the level of economic development. They analyze the World Values Survey data from 33 countries and find that the negative relationship between income inequality and social trust is more pronounced in developed countries, while the relationship is weaker and statistically insignificant in less developed countries. This finding suggests that the contextual factors, such as economic development, may play an essential role in determining the relationship between income inequality and social trust.

The conflicting findings in the literature on the relationship between income inequality and social trust highlight the presence of contextual conflicts in social science research. These conflicts arise due to differences in the research context, such as the level of economic

development and the quality of institutions, which may influence the relationship between the variables of interest.

Understanding the role of contextual factors in shaping the relationship between income inequality and social trust is essential for accurately interpreting the findings of these studies and advancing our knowledge of the underlying mechanisms. Future research should continue to explore the influence of context on the relationship between income inequality and social trust, paying particular attention to the role of economic development, institutional quality, and social welfare policies.

4.2 Case Study 2: Methodological Conflicts: The Impact of Class Size on Student Achievement in Education

The impact of class size on student achievement has been a subject of ongoing debate in educational research. Methodological choices have contributed to conflicting findings, with some studies demonstrating a positive relationship between smaller class sizes and improved student achievement, while others finding little or no significant effect. In this case study, we explore two methodological approaches in the literature on class size and student achievement to illustrate the role of methodological conflict in social science research.

Experimental Approach: Project STAR. One of the most well-known experimental studies in this field is Project STAR (Student/Teacher Achievement Ratio) conducted in Tennessee between 1985 and 1989. This large-scale, randomized controlled trial involved over 11,000 students and aimed to examine the effects of class size on student achievement (Krueger, 1999). The results of this study indicated that students in smaller classes, with 13-17 students, significantly outperformed those in larger classes, with 22-25 students, in both reading and math scores (Krueger, 1999; Mosteller, 1995).

Large-Scale Cross-Sectional Approach. In contrast, Hoxby (2000) conducted a large-scale cross-sectional analysis using data from 649 US elementary schools. She employed a natural experiment approach by exploiting the natural variation in class sizes across schools, resulting from the presence of maximum class size rules. Hoxby (2000) found that class size had little or no significant effect on student achievement after controlling for other school characteristics and teacher quality.

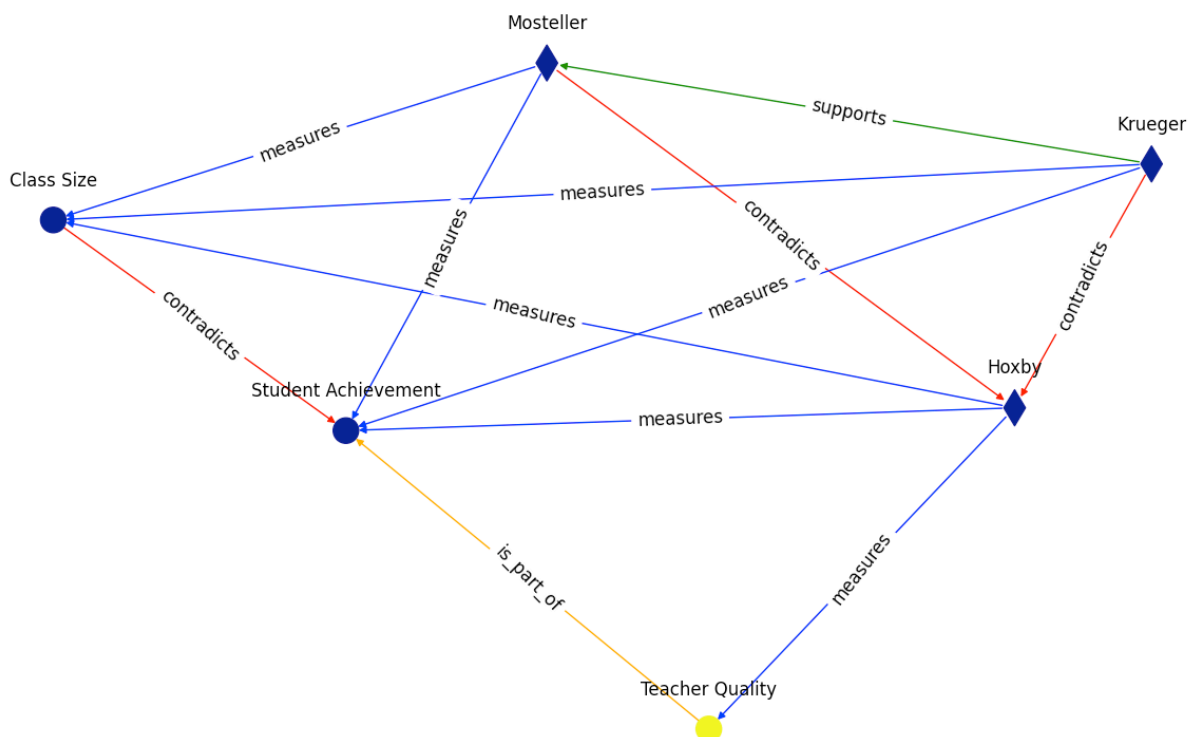


Figure 7. Knowledge Graph: The Impact of Class Size on Student Achievement.

The conflicting findings in the literature on the relationship between class size and student achievement highlight the presence of methodological conflicts in social science research. These conflicts arise due to differences in the research design, data collection, and analysis methods, which may influence the observed relationship between the variables of interest.

Recognizing the role of methodological choices in shaping research outcomes is essential for accurately interpreting the findings of these studies and advancing our understanding of the underlying mechanisms. Future research should continue to explore the impact of class size on

student achievement using diverse methodological approaches, while also considering the influence of other factors, such as teacher quality and school resources.

4.3 Case Study 3: Methodological Conflicts and Sample Size: Social Networks in Social Science

As internet technology becomes increasingly prevalent, massive amounts of data on human interactions can now be obtained without relying on traditional survey and interview methods. It is worth noting that this is not to deny the value of traditional qualitative research methods, but rather to acknowledge that the new methods of internet and big data analysis can more efficiently collect large-scale data that was previously difficult to obtain, revealing the limitations of traditional research methods due to small sample sizes and lower data accuracy.

A key research topic in the emerging field of computational social science is social networks. The advent of various network platforms and real-time communication software has made the analysis of complex social networks possible. A study by Park, Blumenstock, & Macy (2018) published in *Science*, based on data from nearly 60 million internet users, found that long-range ties across large social networks do not weaken with increasing distance and maintain a certain level of strength, playing an important role in the construction of social networks. Prior to this, Granovetter (1973) had already acknowledged that individuals can obtain more new information from weak social ties with acquaintances compared to strong social ties. However, related research suggests that the strength of ties in social networks declines as the degree of social embeddedness decreases. This conclusion has been tested and supported in small-scale social networks (those with fewer nodes and closer distances), but research cannot confirm or refute whether this conclusion applies to long-distance ties. In other words, starting from small-scale data, we can only analyze the conclusion that the strength of relationships weakens as distance increases, without being able to observe the overall

conclusion of social networks. This is not to deny the conclusions of traditional research, but the limitations of small-scale studies constrain their scope. Due to these limitations, the social distances covered by small samples are usually smaller, such as the range of [2,6] indicated by the six degrees of separation theory (Watts & Strogatz, 1998). Large-scale samples, on the other hand, can test social network conditions beyond this range in general, thus providing more general theories and enhancing the understanding of the overall picture in social sciences.

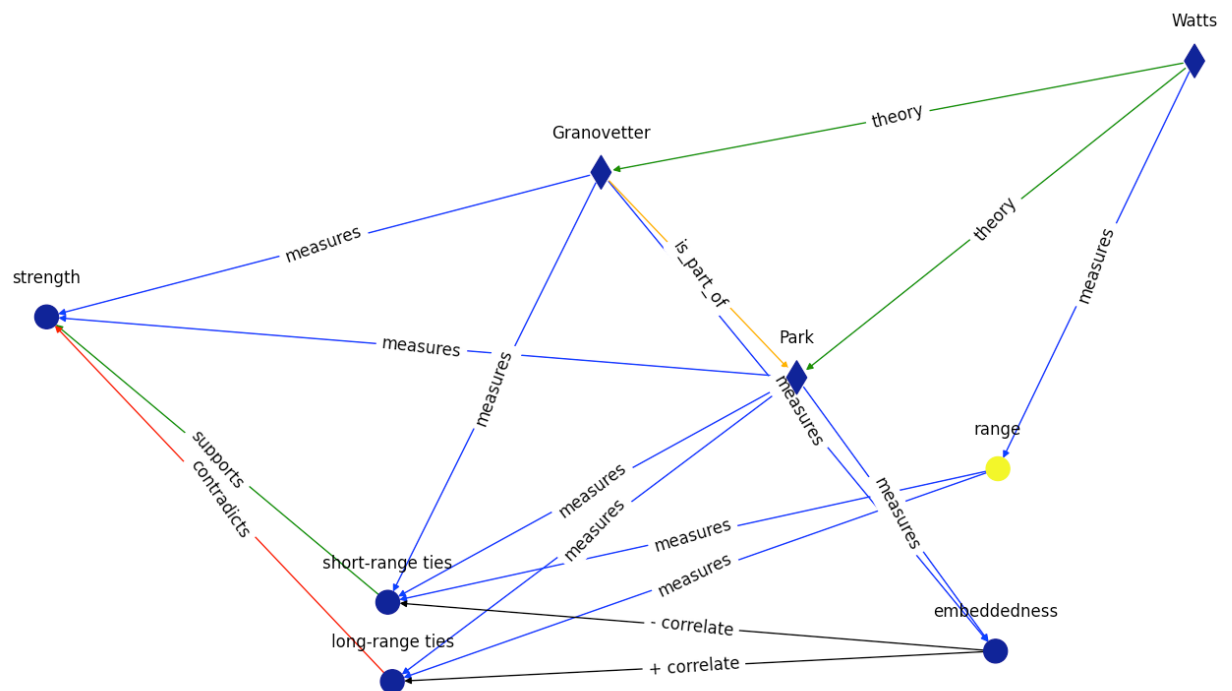


Figure 8. Knowledge Graph: Social Networks.

The difference in samples stems from the differences in measurement capabilities and tools. Granovetter's groundbreaking study in 1973, conducted with only survey questionnaires as a tool, imaginatively analyzed social networks by setting various conditions and achieved reliable results within the sample range. Current social network methods have expanded the boundaries and universality of conclusions due to the emergence of new tools and data sources such as social media. On this issue, Granovetter's (1973) research theory still has great value in explaining strong and weak social ties, but existing empirical research methods can better test and extend the theory.

Furthermore, some issues - difficulties to establish reliable theories and empirical evidence in the past due to a lack of historical data - can now be solved with the support of large-scale samples. A study analyzed how researchers' community networks in a specific research topic reflect research patterns and problems (Kim, Horta, & Jung, 2017); Miller (2011) investigated patterns of mood changes over time; Fujiwara (2018) analyzed the relationship between professors' productivity and promotion over time using public data; and Varga (2019) examined hundreds of thousands of papers, confirming that the distance between disciplines is decreasing and the trend of interdisciplinarity is strengthening. For these issues, we can further interpret them with social science theories, but data science methods provide the empirical basis for these interpretations.

Therefore, the contribution of new research tools to social science research should be recognized. Admittedly, not all issues can be simply addressed with computational sociology methods, and some classic research conclusions and methods still provide guidance for current social science research, serving as a foundation for the development of social sciences. By combining large-scale samples, we can re-validate and re-examine classic social science theories, attempting to build upon and improve the traditional conclusions.

4.4 Case Study 4: Epistemological Conflicts: The Study of Poverty in Social Sciences

The study of poverty in social sciences has been characterized by conflicting epistemological approaches, with positivist and interpretivist researchers adopting different methods and perspectives to understand the complex phenomenon. In this case study, we will examine the key differences between these epistemological approaches in the context of poverty research and discuss the implications of these conflicts on the conclusions drawn from the literature.

Positivist Approach: Quantitative Studies of Poverty. Positivist researchers contend that objective truths can be revealed through empirical research, which often emphasizes the use of quantifiable variables. In the study of poverty, positivist researchers typically rely on measures such as income and consumption to quantify poverty levels (Deaton, 1997). One notable example is the work of Ravallion and Chen (2003), who used household survey data to construct poverty measures based on income and consumption, and subsequently examined the relationship between these measures and economic growth. Their findings suggested that economic growth is a key driver of poverty reduction, highlighting the importance of macroeconomic policies for poverty alleviation.

Interpretivist Approach: Qualitative Studies of Poverty. On the other hand, interpretivist researchers argue that understanding the complex phenomenon of poverty requires a more nuanced approach that emphasizes the importance of qualitative data to reveal the experiences and perspectives of people affected by poverty (Narayan et al., 2000). A prominent example is the Voices of the Poor study, conducted by the World Bank, which used participatory research methods to gather qualitative data from over 20,000 poor individuals across 23 countries (Narayan et al., 2000). This study provided valuable insights into the multidimensional nature of poverty, highlighting the significance of social and political factors in shaping the experiences of poor individuals.

Thus, we can see in the study of poverty, the positivist and interpretivist approaches often yield contrasting perspectives due to their differing epistemological foundations. While the positivist approach might suggest interventions focused on measurable socio-economic factors, i.e., interventions on quantifiable elements like implementing wage regulations, the interpretivist approach might advocate for more holistic, individualized interventions that account for the unique experiences and perspectives of those living in poverty. These fundamentally different conclusions drawn by the two approaches underline the

epistemological conflict in social sciences. The contrasting findings and interpretations in the literature on poverty underscore the presence of epistemological conflicts in social science research. These conflicts arise from differences in researchers' ontological and epistemological beliefs, which inform their methodological choices and their understanding of the causes and consequences of poverty.

Recognizing the importance of diverse perspectives in advancing our understanding of complex social phenomena like poverty is crucial. By embracing a pluralistic approach that combines both quantitative and qualitative methods, researchers can develop a more comprehensive and nuanced understanding of the multidimensional nature of poverty, ultimately informing more effective policies and interventions for poverty alleviation.

4.5 Case Study 5: Epistemological Conflict and Variables: The Bystander Effect in Social Psychology

The bystander effect is a well-established phenomenon in social psychology that demonstrates the influence of context on human behavior. However, research has shown that the generalizability of this effect may be limited by cultural factors, suggesting the presence of epistemological conflicts in the field. In this case study, we will examine the conflicting findings on the bystander effect across cultures and discuss the implications of these inconsistencies for social psychological theory.

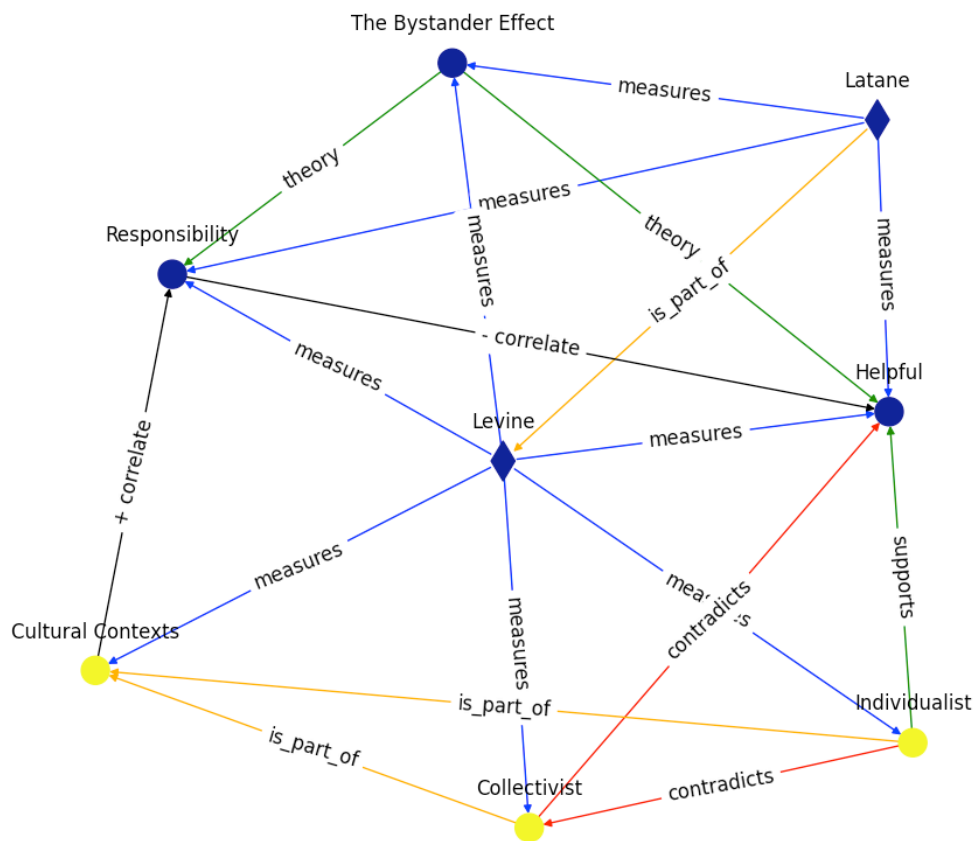


Figure 9. Knowledge Graph: The Bystander Effect in Social Psychology.

The Bystander Effect. The bystander effect refers to the observation that individuals are less likely to intervene in an emergency situation when there are other witnesses present (Latane & Darley, 1968). This phenomenon is thought to arise from a diffusion of responsibility, as individuals assume that others will take action. The classic study by Latane and Darley (1968) demonstrated the bystander effect in a laboratory setting, and subsequent research has replicated these findings in various contexts.

Cultural Variability in the Bystander Effect. Despite the robustness of the bystander effect in individualistic cultures, research has indicated that this phenomenon may not be consistent across cultural contexts. Specifically, studies have found that the bystander effect is less pronounced in collectivist cultures compared to individualist cultures (Levine et al., 2001). In a cross-cultural study, Levine and colleagues (2001) found that individuals from collectivist

cultures were more likely to intervene in emergency situations than their counterparts from individualistic cultures.

This finding suggests that the validity of the bystander effect theory may depend on cultural context, and that cultural factors play an important role in shaping human behavior in emergency situations. These cultural differences may be attributed to variations in social norms, values, and expectations that influence the degree to which individuals feel a sense of responsibility towards others (Triandis, 2018).

The conflicting findings on the bystander effect across cultural contexts highlight the presence of epistemological conflicts in social psychological research⁴. These inconsistencies arise from the influence of the variables involved in the research scope, such as culture, on the generalizability of psychological theories.

Recognizing the importance of context in shaping human behavior is crucial for advancing our understanding of social psychological phenomena. By considering the role of cultural factors in the bystander effect and other social psychological theories, researchers can develop more nuanced and contextually sensitive explanations of human behavior.

⁴ This conflict does involve differences in variables (individualistic vs. collectivist cultures), but this paper believes that the underlying issue is of an epistemological nature. Epistemology concerns the nature, sources, and limits of knowledge. In the case of the bystander effect, researchers have differing beliefs about the conditions under which the effect applies. These differing beliefs stem from disparate understandings about the nature of the knowledge being investigated, for example, is the bystander effect a universally applicable phenomenon, or is it context-dependent and subject to cultural variability? The initial studies on the bystander effect suggested a universal applicability, a view more associated with positivist epistemology. In contrast, subsequent studies showing variations based on cultural factors demonstrated a more interpretivist view, positing that social behaviors are not universal, but are instead shaped by cultural and contextual influences. Hence, this paper argues that the dispute over the bystander effect is not merely about variable selection but also about underlying beliefs on how social psychological knowledge is constructed and understood.

4.6 Case Study 6: Ontological Conflicts and Underlying Assumptions

Regarding the inconsistencies in social science theories on the same problem, Watts (2017) has provided a groundbreaking study that accurately elucidates this phenomenon, and examples have been presented in the introduction of this paper. It appears that there are numerous studies in the existing social science research where, despite addressing almost identical problems, the assumptions and theoretical models constructed are entirely different.

Some of these inconsistencies can be traced back to their origins. Taking the relationship between motivation levels and effort and learning performance in educational psychology as an example, a critique of a social science study based on structural equation modeling pointed out that some research overlooked key mediating variables and proposed erroneous linear models of motivation-learning outcomes (Csizér & Dörnyei, 2005). This linear relationship is inconsistent with many observed empirical findings. Notably, recent experimental research in cognitive science has shown that fitting a Sigmoid function to the relationship between motivation and effort produces better results than linear models (Morris et al., 2020). Liu et al. (2022) also found in their experiment on computer education that the promoting effect of motivation on learning behavior and effectiveness starts to decline when students' motivation reaches a threshold. These findings suggest that greater attention should be paid to the assumptions of variable relationships during empirical research. Statistical significance does not necessarily imply a relationship between two factors.

Even in some influential studies, there is a possibility that erroneous relationship assumptions may render research conclusions less accurate. A highly insightful econometric history study by Acemoglu, Johnson, & Robinson (2001) sought appropriate instrumental variables to test whether institutions significantly promote economic growth. The study used colonial settlers' mortality rates as an instrumental variable, which does not affect institutions and economic conditions, thus addressing potential endogeneity issues and effectively

providing evidence for the original proposition. However, subsequent machine learning work demonstrated that the study's assumption of a linear relationship between institutions and the log of settlers' mortality rates was incorrect, but the strong positive linear relationship between institutions (property rights protection) and economic growth (per-capita GDP) was indeed found (Diallo, 2022). In this set of examples, the theories are eventually consistent, but this does not mean that erroneous assumptions can be ignored. Conversely, Diallo (2022) meaningfully validated the work of Acemoglu, Johnson, & Robinson (2001) using empirical methods, strengthening the reliability of the original research while pointing out its flaws.

Furthermore, some research erroneously interprets correlations as causal relationships. Baert et al. (2019) conducted a study that nicely illustrates this point. The study aimed to investigate whether the relationship between smartphone usage and educational performance is correlational or causal and pointed out that many existing studies have discussed this issue, but their results cannot be interpreted causally due to endogeneity concerns. These results, obtained through linear or logistic regression analyses, may not necessarily indicate the actual relationships, as argued earlier.

In conclusion, it is possible to examine existing empirical research to identify common characteristics of inconsistent theories and seek appropriate solutions or optimization measures for these issues. These examples demonstrate how the validity of social science theories can be influenced by context, methodological choices, underlying assumptions, and the interaction with contextual factors. By carefully examining these conditions, researchers can better understand and address the conflicts that arise in social science theories.

5 Discussions

5.1 Analysis of the Cases and Implications

5.1.1 Identification of Source of Conflicts in Conflicting Theories

Based on the case studies conducted, we are able to identify four types of conflicts that contribute to the discrepancies observed in the literature: epistemological, ontological, methodological, and contextual conflicts (Blaikie, 2007; Creswell & Creswell, 2017).

Epistemological conflicts arise from differences in the fundamental assumptions and beliefs about the nature of knowledge and the methods of acquiring it. For instance, researchers may disagree about the appropriate balance between qualitative and quantitative methods or the role of subjective experience in understanding social phenomena (Guba & Lincoln, 1994).

Ontological conflicts stem from differences in the assumptions about the nature of the social entities or phenomena being studied. For example, researchers may have different beliefs about the role of individuals versus social structures in shaping social outcomes or the existence and relevance of specific psychological constructs (Bhaskar, 2013).

Methodological conflicts arise from differences in the methods and techniques used to study social phenomena. For example, researchers may disagree about the appropriate sampling strategies, data collection techniques, or statistical methods to use in a given context (Maxwell, 2012).

Contextual conflicts emerge from differences in the specific study contexts, such as the populations, settings, or time periods being examined. For example, findings from one study conducted in a particular cultural or historical context may not be directly applicable to another context, leading to inconsistencies and conflicts in the literature (Flyvbjerg, 2006).

By acknowledging and addressing these types of conflicts, researchers can develop a more nuanced understanding of the factors contributing to discrepancies in their findings and work towards building a more coherent body of knowledge.

5.1.2 Factors Contributing to Conflicts and Inconsistencies

As we discussed before, sample size, research context, and variables are key factors that contribute to the inconsistencies in social science research through the literature review we performed in Section 1 and Section 2. The case studies show that these proposed key factors do affect the outcome of the research. The sample size factor is included in the methodological conflicts, the research context is included in the contextual conflicts, and variables factor is included in the epistemological and ontological conflicts.

5.2 Suggestions for Reconciling Inconsistencies and Improving Existing Theories

5.2.1 Proposed Solutions Based on the Identified Patterns

This paper argues that discussions on the sources of conflicts in social science theories identified in this paper should be considered to enhance the quality and reproducibility of social science researches.

Epistemological Conflicts. These conflicts can arise due to the researchers' distinct philosophical stances on knowledge acquisition. For instance, when a theory is supported by positivist research that relies on objective measurements, it may not hold true when examined through an interpretivist lens that prioritizes subjective experiences. Researchers should acknowledge the limitations of their epistemological stance and consider how alternative perspectives may impact the validity of their theories.

Ontological Conflicts. Divergent beliefs about the nature of reality and the entities within it can lead to inconsistencies in the application and understanding of a theory. Researchers should examine the assumptions underpinning their theories and consider the implications of alternative ontological perspectives. This may involve reevaluating the importance of individual agency versus social structures in explaining a phenomenon or exploring the role of unobservable entities like beliefs or values.

Methodological Conflicts. Disagreements in research methods can result in conflicting findings and interpretations. Researchers should be transparent about their methodological choices and consider the potential biases and limitations associated with their chosen methods. To mitigate methodological conflicts, researchers can employ mixed-methods approaches that combine the strengths of quantitative and qualitative techniques or conduct sensitivity analyses to test the robustness of their findings under different methodological assumptions.

Contextual Conflicts. Variations in research settings, populations, or time periods can influence the generalizability of a theory. Researchers should strive to clearly define the context in which their theory is applicable and consider the potential impact of contextual factors on their findings. This may involve conducting cross-cultural or longitudinal studies to test the robustness of a theory in different settings or examining how contextual factors like socioeconomic status, cultural norms, or historical events may influence the relationships under investigation.

In order to ascertain the conditions under which existing theories may be considered valid or invalid, researchers can undertake a comprehensive approach comprising the following stages.

First, a systematic literature review should be conducted to pinpoint the range of studies that either support or contradict a given theory. During this process, particular attention should be paid to the specific contexts and conditions under which these studies were carried out.

Second, by adopting the knowledge graph approach, the relationships between these studies can be analyzed, taking into account various factors such as epistemological, ontological, methodological, and contextual aspects that may contribute to the observed conflicts. This approach facilitates a more in-depth understanding of the connections and potential discrepancies among the studies.

Third, the researcher should identify patterns and trends within the data that indicate the conditions under which the theory holds true. This may include certain cultural settings, specific populations, or particular methodological approaches. Conversely, it is equally important to discern the conditions under which the theory appears to be invalid, such as when certain assumptions or methods are employed, or when specific contextual factors are present.

Finally, the accumulated information should be synthesized in order to refine the theory, elucidating the scope of its applicability and offering guidance for future research. This process aims to circumvent or address potential conflicts and inconsistencies, ultimately strengthening the theoretical foundation and enhancing the overall credibility of the research in question.

5.2.2 Contributions to the Improvement of Social Science Research Methodologies

Based on the results and discussions, we are able to discover several patterns that suggest the conditions under which a theory is valid or invalid.

Consistency Across Contexts. If a theory consistently produces similar results across diverse research settings, populations, and time periods, it suggests that the theory may have broader validity. Conversely, if a theory consistently fails to hold up under different conditions, it may indicate that the theory is invalid or has limited applicability.

Sensitivity to Methodological Choices. If the validity of a theory is highly sensitive to the choice of research methods or the assumptions underlying these methods, it may suggest that the theory is only valid under specific methodological conditions. Researchers should investigate the robustness of their theories under different methodological approaches to identify the conditions under which they are valid or invalid.

Alignment with Underlying Assumptions. Theories that are grounded in well-established epistemological and ontological assumptions are more likely to be valid across a range of conditions. If a theory is found to be valid only when certain assumptions hold, it may

indicate that the theory is invalid or has limited applicability under conditions where those assumptions do not hold. As we stated before, sample size, research context, and variables are important to be considered and examined during the research process.

Interaction with Contextual Factors. If the validity of a theory is influenced by specific contextual factors (e.g., cultural norms, socioeconomic conditions, historical events), it suggests that the theory may be valid only under certain conditions. Researchers should investigate how these factors interact with their theories to identify the conditions under which they are valid or invalid.

This paper believes that researchers can give better applicability and generalization of findings by considering these factors, both in literature study and review as well as in their studies.

5.2.3 Broader Implications for Social Science Research Practices

1. The integration of data science methodologies into social science research has the potential to improve the validity, generalizability, and reproducibility of findings by systematically addressing conflicting theories and uncovering hidden patterns and relationships.
2. The adoption of data-driven approaches can help social scientists to better understand the complexities and nuances of social phenomena, leading to more accurate and comprehensive theories.
3. Encouraging interdisciplinary collaboration and the use of mixed-methods research can facilitate the development of more robust and context-sensitive theories, ultimately enhancing the applicability and credibility of social science research.
4. Incorporating data science methodologies into social science research can also foster a more open and transparent research culture, promoting the sharing of data, methods, and findings, and facilitating the replication and validation of studies.

5. The insights gained from the application of data science methodologies to social science research can contribute to the development of more effective and evidence-based policies and interventions, ultimately benefiting society as a whole.

5.2.4 Why is Reproducibility Important in Social Sciences: Some Reflections on Popper's Falsification

Building upon the aforementioned discussion, the concept of falsification, as proposed by Popper (2005), provides further insight into why reproducibility is a critical issue in scientific research. According to Popper, a theory can only be considered scientific if it can be subjected to the possibility of falsification, which means it must be testable and capable of being refuted by empirical evidence. In this context, reproducibility becomes a key aspect of the scientific process, as it allows for the falsification of a theory based on the consistency of experimental results.

The problem of reproducibility, therefore, poses a significant challenge to the validation of knowledge in both natural and social sciences. When a study's findings cannot be consistently replicated, it casts doubt on the reliability of the initial results and the validity of the underlying theory. Consequently, this hinders the progress of scientific knowledge and undermines the credibility of the research. By embracing Popper's principle of falsification, researchers in both disciplines must strive to ensure the reproducibility of their experiments, allowing for the critical evaluation and potential falsification of theories. In doing so, they contribute to the establishment of robust and reliable scientific knowledge, which ultimately enhances the integrity and progress of their respective fields.

Thus, this paper strongly believes that reproducibility is an important problem in scientific research. This paper suggests the use of falsification ideas to regulate scientific research from a more general perspective, even if this is objectively difficult for the nature of social sciences.

6 Conclusion

6.1 Limitations and Future Research

6.1.1 Limitations of the Current Study

Despite the results achieved in our analysis of conflicting theories in social science using knowledge graphs, there are several limitations that must be acknowledged. First and foremost, our study relies on the quality and representativeness of the collected data. The limited number of papers included in the analysis may not fully capture the breadth and depth of the existing literature in the chosen research areas. Additionally, the data extraction and preprocessing steps may have inadvertently omitted some relevant information or introduced biases, particularly if certain keywords or phrases were not accurately identified.

Another limitation pertains to the construction of the knowledge graph itself. The process of creating nodes, edges, and attributes is largely dependent on the accuracy of the extracted information, which may be susceptible to errors, and it is subjected to decisions about keywords and nodes that are made by researchers. Moreover, the choice of graph representation may not be optimal for capturing the complexity of the relationships between the various entities, particularly if certain relationships have not been adequately represented in the selected studies.

6.1.2 The Potential of Other Data Science Methods

In light of the limitations of the current study, it is worth considering the potential of other data science methodologies to advance our understanding of conflicting theories in social science research. Some promising approaches include:

1. Text mining and natural language processing (NLP): Advanced text mining and NLP techniques could be employed to extract more detailed and accurate information from the literature. For example, topic modeling algorithms such as Latent Dirichlet Allocation (LDA) could be used to identify underlying themes and trends in the

research, while sentiment analysis could provide insights into the polarity of opinions surrounding the conflicting theories.

2. Machine learning and predictive modeling: Supervised and unsupervised machine learning algorithms could be used to predict and classify the relationships between different theories, variables, and studies. For instance, classification algorithms such as Support Vector Machines (SVM) could be used to categorize studies based on their support or contradiction of specific theories, while clustering algorithms like K-means or hierarchical clustering could help identify groups of studies with similar patterns of relationships.
3. Network analysis and graph-based approaches: In addition to the methods used in the current study, other network analysis techniques could be employed to gain a deeper understanding of the knowledge graph. For example, community detection algorithms could be used to identify subgroups of closely related theories, studies, or variables, while link prediction algorithms could help identify potential relationships that have not yet been explicitly documented in the literature.
4. Simulation and agent-based modeling (ABM): Simulation-based approaches, such as agent-based modeling, can provide a valuable means to explore the dynamics of conflicting theories in a controlled, virtual environment. By simulating the interactions between different theories, variables, and studies, it may be possible to identify the factors that contribute to inconsistencies and contradictions and to test the robustness of the identified relationships under various conditions.

While the current study has provided valuable insights into the nature of conflicting theories in social science research, there is significant potential for further exploration using advanced data science methodologies. By leveraging the power of these techniques, we can move closer to understanding the complex interplay of factors that contribute to inconsistencies

in social science theories and ultimately enhance the validity and generalizability of our findings.

6.2 Closing Remarks

This paper draws the following conclusion: there are no significant differences between natural sciences and social sciences in terms of empirical research methods. Although social sciences face more difficulties in experimental conditions compared to natural sciences, the issue of reproducibility is not a key characteristic that distinguishes the two disciplines, as it is a common problem in all sciences. Considering that social sciences study not only objective facts but also subjective aspects, both qualitative and quantitative research methods are required to support their investigations, while natural sciences do not need to consider qualitative research and issues of meaning. In light of the challenges in experimental conditions for social sciences, and the fact that qualitative research should have different evaluation criteria from quantitative research, the application of data science methods in social sciences can help enhance research reliability.

Existing research suggests that although big data has shown a significant impact on social sciences, many large-scale analyses related to social sciences are still conducted by computer scientists, and this influence has not been reflected in top-tier sociology journals, such as the *American Journal of Sociology* and *American Sociological Review* (Lazer & Radford, 2017). This indicates that integrating data science methods with existing social science research frameworks still requires considerable effort.

This paper posits that more computational social science research will contribute to the further development of empirical social science research in addressing issues of inconsistency and reproducibility. At the same time, researchers should remain mindful that the role of data science in social science research is to support the development and testing of social science

theories. Data should not dictate research 100%; reasonable and qualitative interpretations of data are equally important. Furthermore, the best way to ultimately verify the effectiveness of a theory is still to apply it in real-world situations and solve practical problems.

Ethical Statement: No human subjects or animals are involved in this research.

Conflicts of Interest: The author declares no potential conflicts of interest.

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References

- Acemoglu, D., Johnson, S. and Robinson, J.A., 2001. The colonial origins of comparative development: An empirical investigation. *American economic review*, 91(5), pp.1369-1401.
- Antona, M. and Babin, D., 2001. Multiple interest accommodation in African forest management projects: between pragmatism and theoretical coherence. *International Journal of Agricultural Resources, Governance and Ecology*, 1(3-4), pp.286-305.
- Arbuckle, B.S., Kansa, S.W., Kansa, E., Orton, D., Çakırlar, C., Gourichon, L., Atici, L., Galik, A., Marciniak, A., Mulville, J. and Buitenhuis, H., 2014. Data sharing reveals complexity in the westward spread of domestic animals across Neolithic Turkey. *PloS one*, 9(6), p.e99845.
- Axelrod, R., 1997. Advancing the art of simulation in the social sciences. In *Simulating social phenomena* (pp. 21-40). Springer Berlin Heidelberg.
- Baert, S., Vujić, S., Amez, S., Claeskens, M., Daman, T., Maeckelberghe, A., Omeij, E. and De Marez, L., 2020. Smartphone use and academic performance: correlation or causal relationship?. *Kyklos*, 73(1), pp.22-46.
- Baker, M., 2016. 1,500 scientists lift the lid on reproducibility. *Nature*, 533(7604).
- Baranowski, T., Lin, L.S., Wetter, D.W., Resnicow, K. and Hearn, M.D., 1997. Theory as mediating variables: Why aren't community interventions working as desired?. *Annals of Epidemiology*, 7(7), pp.S89-S95.
- Bhaskar, R., 2013. *A realist theory of science*. Routledge.
- Bickle, J., 2007. Ruthless reductionism and social cognition. *Journal of Physiology-Paris*, 101(4-6), pp.230-235.
- Biehl, J., 2013. Ethnography in the Way of Theory. *Cultural Anthropology*, 28(4), pp.573-597.

- Bishop, C.M. and Nasrabadi, N.M., 2006. *Pattern recognition and machine learning* (Vol. 4, No. 4, p. 738). New York: springer.
- Blaikie, N., 2007. *Approaches to social enquiry: Advancing knowledge*. Polity.
- Boiché, J., Sarrazin, P.G., Grouzet, F.M., Pelletier, L.G. and Chanal, J.P., 2008. Students' motivational profiles and achievement outcomes in physical education: A self-determination perspective. *Journal of educational psychology*, 100(3), p.688.
- Bollen, K.A., 1989. *Structural equations with latent variables* (Vol. 210). John Wiley & Sons.
- Borenstein, M., Hedges, L.V., Higgins, J.P. and Rothstein, H.R., 2021. *Introduction to meta-analysis*. John Wiley & Sons.
- Bronfenbrenner, U., 1979. *The ecology of human development: Experiments by nature and design*. Harvard university press.
- Button, K.S., Ioannidis, J.P., Mokrysz, C., Nosek, B.A., Flint, J., Robinson, E.S. and Munafò, M.R., 2013. Power failure: why small sample size undermines the reliability of neuroscience. *Nature reviews neuroscience*, 14(5), pp.365-376.
- Cerasoli, C.P., Nicklin, J.M. and Ford, M.T., 2014. Intrinsic motivation and extrinsic incentives jointly predict performance: a 40-year meta-analysis. *Psychological bulletin*, 140(4), p.980.
- Chen, N.C., Drouhard, M., Kocielnik, R., Suh, J. and Aragon, C.R., 2018. Using machine learning to support qualitative coding in social science: Shifting the focus to ambiguity. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 8(2), pp.1-20.
- Chen, B., Liu, K., Liu, X., Shi, P., Lin, C. and Zhang, H., 2013. Approximation-based adaptive neural control design for a class of nonlinear systems. *IEEE transactions on cybernetics*, 44(5), pp.610-619.
- Chen, P., Lu, Y., Zheng, V.W., Chen, X. and Yang, B., 2018. Knowedu: A system to construct knowledge graph for education. *Ieee Access*, 6, pp.31553-31563.

- Clunas, C., 1999. Modernity global and local: Consumption and the rise of the West. *The American Historical Review*, 104(5), pp.1497-1511.
- Clonts, J.G., 1992. The Concept of Reliability as It Pertains to Data from Qualitative Studies.
- Creswell, J.W. and Creswell, J.D., 2017. *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage publications.
- Croitoru, M., Marquis, P., Rudolph, S. and Stapleton, G. eds., 2018. *Graph Structures for Knowledge Representation and Reasoning: 5th International Workshop, GKR 2017, Melbourne, VIC, Australia, August 21, 2017, Revised Selected Papers* (Vol. 10775). Springer.
- Csizér, K. and Dörnyei, Z., 2005. The internal structure of language learning motivation and its relationship with language choice and learning effort. *The modern language journal*, 89(1), pp.19-36.
- Davidson, E., Edwards, R., Jamieson, L. and Weller, S., 2019. Big data, qualitative style: a breadth-and-depth method for working with large amounts of secondary qualitative data. *Quality & quantity*, 53, pp.363-376.
- de Haan, R., Tiddi, I. and Beek, W., 2021. Discovering research hypotheses in social science using knowledge graph embeddings. In *The Semantic Web: 18th International Conference, ESWC 2021, Virtual Event, June 6–10, 2021, Proceedings 18* (pp. 477-494). Springer International Publishing.
- Deaton, A., 1997. *The analysis of household surveys: a microeconomic approach to development policy*. World Bank Publications.
- Diallo, B., 2022. Machine learning approaches to testing institutional hypotheses: the case of Acemoglu, Johnson, and Robinson (2001). *Empirical Economics*, 62(5), pp.2587-2600.

- Dou, J., Qin, J., Jin, Z. and Li, Z., 2018. Knowledge graph based on domain ontology and natural language processing technology for Chinese intangible cultural heritage. *Journal of Visual Languages & Computing*, 48, pp.19-28.
- Du, M., Liu, N. and Hu, X., 2019. Techniques for interpretable machine learning. *Communications of the ACM*, 63(1), pp.68-77.
- Dunning, T., 2012. *Natural experiments in the social sciences: a design-based approach*. Cambridge University Press.
- Ehrlinger, L. and Wöß, W., 2016. Towards a definition of knowledge graphs. *SEMANTiCS (Posters, Demos, SuCCESS)*, 48(1-4), p.2.
- Elgar, F.J. and Aitken, N., 2011. Income inequality, trust and homicide in 33 countries. *European Journal of Public Health*, 21(2), pp.241-246.
- Epstein, J.M. and Axtell, R., 1996. *Growing artificial societies: social science from the bottom up*. Brookings Institution Press.
- Flyvbjerg, B., 2006. Five misunderstandings about case-study research. *Qualitative inquiry*, 12(2), pp.219-245.
- Fujiwara, A., 2018. When can academic researchers rest? An event history analysis on researchers' research productivity and promotion in academia from 1980 to 2016 in Japan. *Open Review of Educational Research*, 5(1), pp.130-143.
- Gandomi, A. and Haider, M., 2015. Beyond the hype: Big data concepts, methods, and analytics. *International journal of information management*, 35(2), pp.137-144.
- Giddens, A., 1986a. *Sociology: A Brief but Critical Introduction*. Bloomsbury Publishing.
- Giddens, A., 1986b. *The constitution of society: Outline of the theory of structuration* (Vol. 349). Univ of California Press.
- González-Bailón, S., 2013. Social science in the era of big data. *Policy & internet*, 5(2), pp.147-160.

- Granovetter, M.S., 1973. The strength of weak ties. *American journal of sociology*, 78(6), pp.1360-1380.
- Grimmer, J., Roberts, M.E. and Stewart, B.M., 2021. Machine learning for social science: An agnostic approach. *Annual Review of Political Science*, 24, pp.395-419.
- Guba, E.G. and Lincoln, Y.S., 1994. Competing paradigms in qualitative research. *Handbook of qualitative research*, 2(163-194), p.105.
- Hogan, A., Blomqvist, E., Cochez, M., d'Amato, C., Melo, G.D., Gutierrez, C., Kirrane, S., Gayo, J.E.L., Navigli, R., Neumaier, S. and Ngomo, A.C.N., 2021. Knowledge graphs. *ACM Computing Surveys (CSUR)*, 54(4), pp.1-37.
- Howick, J., Koletsi, D., Ioannidis, J.P., ... and Johnson, T., 2022. Most healthcare interventions tested in Cochrane Reviews are not effective according to high quality evidence: a systematic review and meta-analysis. *Journal of clinical epidemiology*.
- Hoxby, C.M., 2000. The effects of class size on student achievement: New evidence from population variation. *The Quarterly Journal of Economics*, 115(4), pp.1239-1285.
- Ioannidis, J.P., 2005. Why most published research findings are false. *PLoS medicine*, 2(8), p.e124.
- Ioannidis, J.P., 2012. Why science is not necessarily self-correcting. *Perspectives on Psychological Science*, 7(6), pp.645-654.
- Ioannidis, J. and Doucouliagos, C., 2013. What's to know about the credibility of empirical economics?. *Journal of Economic Surveys*, 27(5), pp.997-1004.
- Jackson, J.C., Rand, D., Lewis, K., Norton, M.I. and Gray, K., 2017. Agent-based modeling: A guide for social psychologists. *Social Psychological and Personality Science*, 8(4), pp.387-395.
- Keohane, R.O. and Nye, J.S., 1977. Power and interdependence: world politics in transition (Boston, Little and Brown).

- Keynes, J.M., 1937. The general theory of employment. *The quarterly journal of economics*, 51(2), pp.209-223.
- Krueger, A.B., 1999. Experimental estimates of education production functions. *The quarterly journal of economics*, 114(2), pp.497-532.
- Kusurkar, R.A., Ten Cate, T.J., Vos, C.M.P., Westers, P. and Croiset, G., 2013. How motivation affects academic performance: a structural equation modelling analysis. *Advances in health sciences education*, 18(1), pp.57-69.
- Knorr-Cetina, K.D., 1981. Social and scientific method or what do we make of the distinction between the natural and the social sciences?. *Philosophy of the Social Sciences*, 11(3), pp.335-359.
- Ladyman, J., Lambert, J. and Wiesner, K., 2013. What is a complex system?. *European Journal for Philosophy of Science*, 3(1), pp.33-67.
- Lakatos, I., 1976. Falsification and the methodology of scientific research programmes. In *Can theories be refuted?* (pp. 205-259). Springer, Dordrecht.
- Larsson, S., 2009. A pluralist view of generalization in qualitative research. *International journal of research & method in education*, 32(1), pp.25-38.
- Latane, B. and Darley, J.M., 1968. Group inhibition of bystander intervention in emergencies. *Journal of personality and social psychology*, 10(3), p.215.
- Lawless, W.F., 2017. The entangled nature of interdependence. Bistability, irreproducibility and uncertainty. *Journal of Mathematical Psychology*, 78, pp.51-64.
- Lazer, D., Pentland, A., Adamic, L., Aral, S., Barabasi, A.L., Brewer, D., Christakis, N., Contractor, N., Fowler, J., Gutmann, M. and Jebara, T., 2009. Social science. Computational social science. *Science (New York, NY)*, 323(5915), pp.721-723.

- Lazer, D.M., Pentland, A., Watts, D.J., Aral, S., Athey, S., Contractor, N., Freelon, D., Gonzalez-Bailon, S., King, G., Margetts, H. and Nelson, A., 2020. Computational social science: Obstacles and opportunities. *Science*, 369(6507), pp.1060-1062.
- Lazer, D. and Radford, J., 2017. Data ex machina: introduction to big data. *Annual Review of Sociology*, 43, pp.19-39.
- Leamer, E.E., 1998. Let's Take the Con Out of Econometrics, *American Economic Review*, 73(1), March, 31-43.
- Levens, E., 2021. A mass cytometry application of a community structure generating algorithm derived from a combination of Leiden and Girvan-Newman methods.
- Levine, R.V., Norenzayan, A. and Philbrick, K., 2001. Cross-cultural differences in helping strangers. *Journal of cross-cultural psychology*, 32(5), pp.543-560.
- Little, D., 1993. On the scope and limits of generalizations in the social sciences. *Synthese*, 97, pp.183-207.
- Liu, H., Wu, Z., Lu, Y. and Zhu, L., 2022. Exploring the Balance Between Computational Thinking and Learning Motivation in Elementary Programming Education: An Empirical Study with Game-based Learning. *IEEE Transactions on Games*, 15(1), pp.95-107.
- Lorenz, E.N. and Haman, K., 1996. The essence of chaos. *Pure and Applied Geophysics*, 147(3), pp.598-599.
- Louth, J., 2011. From Newton To Newtonianism: Reductionism And The Development Of The Social Sciences. *Emergence: Complexity & Organization*, 13(4).
- Malik, M.M., 2020. A hierarchy of limitations in machine learning. *arXiv preprint arXiv:2002.05193*.
- Marx, K. and Engels, F., 1848. The Communist Manifesto, intro. E. Hobsbawm. *London and New York: Verso*.
- Maxwell, J.A., 2012. *Qualitative research design: An interactive approach*. Sage publications.

- Merton, Robert King. *Social theory and social structure*. Simon and Schuster, 1968.
- Miller, G., 2011. Social Scientists Wade Into the Tweet Stream. *Science*, 333(6051), p.1814.
- Mills, K.A., 2018. What are the threats and potentials of big data for qualitative research?. *Qualitative Research*, 18(6), pp.591-603.
- Mills, K.A., 2019. *Big data for qualitative research* (p. 90). Taylor & Francis.
- Morgenthau, H.J., Thompson, K.W. and Clinton, W.D., 1985. Politics among nations: The struggle for power and peace.
- Morris, L.S., Norbury, A., Smith, D.A., Harrison, N.A., Voon, V. and Murrough, J.W., 2020. Dissociating self-generated volition from externally-generated motivation. *PloS one*, 15(5), p.e0232949.
- Mosteller, F., 1995. The Tennessee study of class size in the early school grades. *The future of children*, pp.113-127.
- Müller, P. and Roeder, K., 1997. A Bayesian semiparametric model for case-control studies with errors in variables. *Biometrika*, 84(3), pp.523-537.
- Murayama, K., Matsumoto, M., Izuma, K. and Matsumoto, K., 2010. Neural basis of the undermining effect of monetary reward on intrinsic motivation. *Proceedings of the National Academy of Sciences*, 107(49), pp.20911-20916.
- Narayan, D., Patel, R., Schafft, K., Rademacher, A. and Koch-Schulte, S., 2000. Voices of the Poor: Can Anyone Hear Us?
- Park, P.S., Blumenstock, J.E. and Macy, M.W., 2018. The strength of long-range ties in population-scale social networks. *Science*, 362(6421), pp.1410-1413.
- Parsons, T., 1952. The superego and the theory of social systems. *Psychiatry*, 15(1), pp.15-25.
- Paulheim, H., 2017. Knowledge graph refinement: A survey of approaches and evaluation methods. *Semantic web*, 8(3), pp.489-508.
- Peters, U., 2020. Values in science: assessing the case for mixed claims. *Inquiry*, pp.1-12.

- Petticrew, M. and Roberts, H., 2008. *Systematic reviews in the social sciences: A practical guide*. John Wiley & Sons.
- Plomin, R., 2018. *Blueprint: How DNA Makes Us Who We Are*. London. Allen Lane.
- Pluchino, A., Biondo, A.E. and Rapisarda, A., 2018. Talent versus luck: The role of randomness in success and failure. *Advances in Complex systems*, 21(03n04), p.1850014.
- Pope, C. and Mays, N., 1995. Qualitative research: reaching the parts other methods cannot reach: an introduction to qualitative methods in health and health services research. *bmj*, 311(6996), pp.42-45.
- Popper, K., 2005. *The logic of scientific discovery*. Routledge.
- Raftery, A.E., 2001. Statistics in sociology, 1950–2000: A selective review. *Sociological Methodology*, 31(1), pp.1-45.
- Rahwan, I., Cebrian, M., Obradovich, N., Bongard, J., Bonnefon, J.F., Breazeal, C., Crandall, J.W., Christakis, N.A., Couzin, I.D., Jackson, M.O. and Jennings, N.R., 2022. Machine behaviour. *Machine Learning and the City: Applications in Architecture and Urban Design*, pp.143-166.
- Ravallion, M. and Chen, S., 2003. Measuring pro-poor growth. *Economics letters*, 78(1), pp.93-99.
- Resnik, D.B. and Shamoo, A.E., 2017. Reproducibility and research integrity. *Accountability in research*, 24(2), pp.116-123.
- Rothstein, B. and Uslaner, E.M., 2005. All for all: Equality, corruption, and social trust. *World politics*, 58(1), pp.41-72.
- Salganik, M.J., 2019. *Bit by bit: Social research in the digital age*. Princeton University Press.
- Sawyer, R.K., 2004. The mechanisms of emergence. *Philosophy of the social sciences*, 34(2), pp.260-282.

- Schulze, G.E., 1792. Aenesidemus, or Concerning the Foundations of the Philosophy of the Elements Issued by Prof. Reinhold in Jena Together with a Defence of Skepticism Against the Pretensions of The Critique of Reason.”. *Between Kant and Hegel: Texts in the Development of Post-Kantian Idealism*, pp.104-135.
- Schwarz, G., 1978. Estimating the dimension of a model. *The annals of statistics*, pp.461-464.
- Schwinger, M., Steinmayr, R. and Spinath, B., 2009. How do motivational regulation strategies affect achievement: Mediated by effort management and moderated by intelligence. *Learning and individual differences*, 19(4), pp.621-627.
- Sidi, Y. and Harel, O., 2018. The treatment of incomplete data: reporting, analysis, reproducibility, and replicability. *Social Science & Medicine*, 209, pp.169-173.
- Silverman, D., 1998. Qualitative research: meanings or practices?. *Information systems journal*, 8(1), pp.3-20.
- Smaldino, P.E. and McElreath, R., 2016. The natural selection of bad science. *Royal Society open science*, 3(9), p.160384.
- Smith, A., 1776. An inquiry into the nature and causes of the wealth of nations: Volume One. London: printed for W. Strahan; and T. Cadell, 1776.
- Song, M., Zhang, J., Liu, X., Zhang, L., Hao, X. and Li, M., 2023. Developments and Trends in Energy Poverty Research—Literature Visualization Analysis Based on CiteSpace. *Sustainability*, 15(3), p.2576.
- Stamou, F., 2022. Data science for social good: development of a knowledge graph targeted to sustainable development goals.
- Stanley, T.D. and Doucouliagos, H., 2012. *Meta-regression analysis in economics and business*. routledge.

- Sturman, M.C., Sturman, A.J. and Sturman, C.J., 2022. Uncontrolled control variables: The extent that a researcher's degrees of freedom with control variables increases various types of statistical errors. *Journal of Applied Psychology*, 107(1), p.9.
- Sui, D. and Kedron, P., 2021. Reproducibility and Replicability in the Context of the Contested Identities of Geography. *Annals of the American Association of Geographers*, 111(5), pp.1275-1283.
- Triandis, H.C., 2018. *Individualism and collectivism*. Routledge.
- Troost, C., Huber, R., Bell, A.R., van Delden, H., Filatova, T., Le, Q.B., Lippe, M., Niamir, L., Polhill, J.G., Sun, Z. and Berger, T., 2023. How to keep it adequate: A protocol for ensuring validity in agent-based simulation. *Environmental Modelling & Software*, 159, p.105559.
- Uslaner, E.M., 2002. The moral foundations of trust. *Available at SSRN 824504*.
- Varga, A., 2019. Shorter distances between papers over time are due to more cross-field references and increased citation rate to higher-impact papers. *Proceedings of the National Academy of Sciences*, 116(44), pp.22094-22099.
- Verschuren, P.J., 2001. Holism versus reductionism in modern social science research. *Quality and Quantity*, 35, pp.389-405.
- Wasserman, S. and Faust, K., 1994. *Social network analysis: Methods and applications*.
- Watts, D.J., 2017. Should social science be more solution-oriented?. *Nature Human Behaviour*, 1(1), pp.1-5.
- Watts, D.J. and Strogatz, S.H., 1998. Collective dynamics of 'small-world' networks. *nature*, 393(6684), pp.440-442.
- Weinberg, G.M., 1975. An introduction to general systems thinking. *Wiley Series on Systems Engineering and Analysis*.

- Xia, T. and Gu, Y., 2019, July. Building terrorist knowledge graph from global terrorism database and wikipedia. In *2019 IEEE International Conference on Intelligence and Security Informatics (ISI)* (pp. 194-196). IEEE.
- Yun, H., He, Y., Lin, L., Pan, Z. and Zhang, X., 2019. Construction research and application of poverty alleviation knowledge graph. In *Web Information Systems and Applications: 16th International Conference, WISA 2019, Qingdao, China, September 20-22, 2019, Proceedings 16* (pp. 430-442). Springer International Publishing.
- Zhang, J., Wang, W., Xia, F., Lin, Y.R. and Tong, H., 2020. Data-driven computational social science: A survey. *Big Data Research*, 21, p.100145.